NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

16 June 2026

258 verified · 472 raw · 5 sources

258	0	258	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	236	Media
TechCrunch AI	8	Media
HackerNews	7	Community
The Verge AI	5	Media
MIT Tech Review	2	Media

CONFIRMED SIGNALS — 258 stories

CONF

Show HN: I wrote a C++ ray tracer from scratch without AI

HackerNews 60% HN 148

No summary — see link for full article.

https://github.com/themartiano/luz

CONF

Show HN: Claude Code for Visual Studio (native diff with accept/reject)

HackerNews 50% HN 19

VS Code and JetBrains have official Claude Code IDE integration, but Visual Studio doesn't. There's an open GitHub issue with a lot of +1s so I built it myself (check out the gif in the GitHub repo for a visual example). It implements the same protocol the official plugins use,...

https://github.com/firish/claude_code_vs

CONF

Show HN: I used Claude Mythos to build my startup in 1 day

HackerNews 50% HN 9

This sounds clickbait, but it’s true: I used Claude Mythos to build the full site in 1 day. Then Anthropic removed the model, so I had to go back to Opus.

https://www.brandlm.ai/

CONF

Show HN: When Will AI? – A timeline of top AI predictions

HackerNews 50% HN 6

No summary — see link for full article.

https://whenwill.ai

CONF

Show HN: Memorypad – A note editor for your daily notes, in Markdown

HackerNews 50% HN 6

No summary — see link for full article.

https://memorypad.io/

CONF

Show HN: Transpilatron – an AI tool that converts Python code into C binaries

HackerNews 50% HN 5

No summary — see link for full article.

https://github.com/NoodlixProject/transpilatron

CONF

Show HN: Micro Coach – an AI workout planner built by a former personal trainer

HackerNews 50% HN 3

I spent the past year building Micro Coach with two partners. It creates personalized 4-week training blocks based on your goals, schedule, experience level, equipment, and preferences. Users can also generate individual workouts, track progress, and view training analytics. I...

https://microcoachapp.com/

CONF

Malaysia’s AI agent-powered messaging app Respond.io raises $62.5M, eyes acquisitions

TechCrunch AI 50%

Respond.io, one of Malaysia startups to watch, uses AI agents to handle high volumes of customer inquiries and charges per convo, not per seat.

https://techcrunch.com/2026/06/15/malaysias-respond-io-raises-62-5m-eyes-acquisitions-i...

CONF

Sundar Pichai faces boos, walkout at Stanford graduation ceremony over Google's Israel, ICE ties

TechCrunch AI 50%

AI is once again at the heart of a college graduation protest — this time for the technology's use in Google's defense contracts.

https://techcrunch.com/2026/06/15/sundar-pichai-faces-boos-walkout-at-stanford-graduati...

0 CONF

The US government's Anthropic models ban was never about an AI jailbreak

TechCrunch AI 50%

The Trump administration's decision that forced Anthropic to pull its latest cybersecurity models could be reactionary, retaliatory, or both, but the message is clear: The AI industry isn't immune from U.S. government interference.

https://techcrunch.com/2026/06/15/the-us-governments-anthropic-models-ban-was-never-abo...

1 CONF

Meta's new 'AI Mode' on Facebook pulls from public info across its platforms

TechCrunch AI 50%

Meta announced Monday that it's rolling out a wave of new AI features on Facebook, the latest sign of the company's effort to catch up in the AI race and keep users more engaged on the platform.

https://techcrunch.com/2026/06/15/metas-new-ai-mode-on-facebook-pulls-from-public-info-...

2 CONF

Cybersecurity vets protest 'dangerous' US government ban on Anthropic's most powerful models

TechCrunch AI 50%

A group made up of dozens of cybersecurity experts urged the White House to remove export-control restrictions on Anthropic's Fable and Mythos models, arguing that the order is going to limit the ability of cybersecurity defenders to secure their software and products.

https://techcrunch.com/2026/06/15/cybersecurity-vets-protest-dangerous-us-government-ba...

3 CONF

Salesforce acquires AI customer service platform Fin for $3.6B

TechCrunch AI 50%

Salesforce says it wants to use Fin's team and technology to improve Agentforce, its existing enterprise platform that businesses can use to build custom AI agents that automate tasks.

https://techcrunch.com/2026/06/15/salesforce-acquires-ai-customer-service-platform-fin-...

4 CONF

Sarvam becomes India's newest AI unicorn with $234 million funding round led by HCLTech

TechCrunch AI 50%

Indian IT services company HCLTech is investing $150 million in the Bengaluru startup.

https://techcrunch.com/2026/06/15/sarvam-becomes-indias-newest-ai-unicorn-with-234-mill...

5 CONF

As AI agents become employees, NewCore emerges with $66M to give them identities

TechCrunch AI 50%

NewCore argues the next challenge in enterprise security will be managing AI agents, not people.

https://techcrunch.com/2026/06/15/ai-agents-are-becoming-employees-newcore-emerges-with...

6 **CONF**

Why do South Koreans love AI so much?

MIT Tech Review **50%**

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. When I landed in Seoul after a grueling 12-hour flight from San Francisco, I walked through an unmanned immigration checkpoint, where a...

https://www.technologyreview.com/2026/06/15/1138983/why-do-south-koreans-love-ai-so-much/

7 **CONF**

This man with ALS is “the first power user” of a brain implant that lets him speak

MIT Tech Review **50%**

Casey Harrell has had a set of electrodes embedded in his brain for almost three years. Harrell, who has amyotrophic lateral sclerosis (ALS) and is paralyzed, first used his brain-computer interface (BCI) to “speak” sentences with the help of a research team in 2023. Since then,...

https://www.technologyreview.com/2026/06/15/1138953/man-als-first-power-user-brain-impl...

8 **CONF**

Inside the fight over Claude Mythos 5

The Verge AI **50%**

As the rest of the country celebrated the USA's first World Cup win and the New York Knicks championship, Anthropic spent its weekend fighting the Trump administration over its latest model release. At 5:21 PM on Friday, the company received a US export control directive to...

https://www.theverge.com/ai-artificial-intelligence/950412/anthropic-trump-adminstratio...

9 **CONF**

Facebook’s new AI Mode search gets its info from public posts

The Verge AI **50%**

Your public Facebook posts could help inform AI-generated results in Meta's new AI Mode. When you search on Facebook, the "AI Mode" option will appear alongside the usual search modes like "People" and "Marketplace." It's one of several new AI features Meta is rolling out...

https://www.theverge.com/tech/950264/meta-ai-mode-search-facebook

0 **CONF**

All the news about Anthropic’s new AI fight with the White House

The Verge AI **50%**

Anthropic was already navigating one dispute with the government in its standoff with the Pentagon, and then came an order on June 12th to block off foreign access to its most recently released AI models, Fable 5 and Mythos 5. When they launched on June 9th, Anthropic said...

https://www.theverge.com/ai-artificial-intelligence/950026/anthropic-fable-mythos-ban-a...

1 **CONF**

Trump’s Anthropic shutdown just made the case for non-American AI

The Verge AI **50%**

At Washington's request, Anthropic suddenly took its newest and most powerful AI models offline over the weekend. The American company said it had little choice after the White House demanded it block access for all foreign nationals, including its own employees. Abroad, the...

https://www.theverge.com/ai-artificial-intelligence/949986/anthropic-fable-mythos-shutd...

2 **CONF**

Big Tech’s desperate last push at AI regulation

The Verge AI **50%**

For months, Big Tech's Washington lobbyists have chased after the holy grail of pro-AI legislation: preemption. This would be a comprehensive federal law, passed in Congress and signed by the president, applying one set of AI rules across the entire country and overriding the...

https://www.theverge.com/policy/949970/ai-regulation-child-safety-kosa-congress

3 **CONF**

A Definition of Good Explanations and the Challenges Explaining LLM Outputs

ArXiv CS.AI **50%**

arXiv:2606.14838v1 Announce Type: new Abstract: How to define a good explanation is a long-standing philosophical debate which has found recent renewed interest in the context of AI outputs. Explainability is crucial for AI adoption in many contexts, but in order to produce good...

https://arxiv.org/abs/2606.14838

4 **CONF**

Optimising Temporary Accommodation Placement Across London with AI-Powered SaaS in E-Governance Systems

ArXiv CS.AI **50%**

arXiv:2606.16652v1 Announce Type: cross Abstract: Temporary accommodation has become a major fiscal and administrative pressure for English local authorities, particularly in London, where demand and costs have risen sharply. This paper documents the creation and use of DOMUS, a...

https://arxiv.org/abs/2606.16652

5 **CONF**

ToolMenuBench: Benchmarking Tool-Menu Filtering Strategies for Reliable and Efficient LLM Agents

ArXiv CS.AI **50%**

arXiv:2606.15508v1 Announce Type: new Abstract: Tool-augmented large language model agents increasingly operate over large tool libraries, but existing evaluations often focus on whether a model can call a tool correctly rather than how the visible tool menu shapes reliability,...

https://arxiv.org/abs/2606.15508

6 **CONF**

A Causal Model of Theory of Mind in Conflict for Artificial Intelligence

ArXiv CS.AI **50%**

arXiv:2606.16944v1 Announce Type: new Abstract: Theory of mind (ToM), the capacity to ascribe mental states to others and use those ascriptions for prediction and inference, is widely assumed to be essential for effective human-machine integration. Existing AI-ToM models address...

https://arxiv.org/abs/2606.16944

7 **CONF**

Metric Match: A Subset Selection Approach to Evaluating LLM Judge Reliability

ArXiv CS.AI **50%**

arXiv:2606.15029v1 Announce Type: new Abstract: LLM judges are used to reduce the need for costly human labor in evaluating open-ended text generation. However, the reliability of these judges depends critically on their alignment with human raters -- a property that itself...

https://arxiv.org/abs/2606.15029

8 **CONF**

AuAu: A Benchmark for Auditing Authoritarian Alignment in Large Language Models

ArXiv CS.AI **50%**

arXiv:2606.16127v1 Announce Type: cross Abstract: The worldwide surge of authoritarianism, combined with the increasing central role in users' everyday lives, raises the question of to what extent specific models exhibit or promote authoritarian attitudes and characteristics. We...

https://arxiv.org/abs/2606.16127

9 **CONF**

Fusion is not one-size-fits-all: Cross-Modal Representation Alignment for Time-to-Event Modeling

ArXiv CS.AI **50%**

arXiv:2606.15038v1 Announce Type: new Abstract: Accurate time-to-event (TTE) prediction from multimodal clinical data remains challenging due to modality imbalance and distribution shift. We introduce a foundation model-driven framework for cross-modal representation alignment...

https://arxiv.org/abs/2606.15038

0 **CONF**

Risk-Aware LLM Agents for Geospatial Data Retrieval: Design and Preliminary Adversarial Evaluation

ArXiv CS.AI **50%**

arXiv:2606.15077v1 Announce Type: new Abstract: We present an LLM-driven framework for retrieving remote sensing data from cloud-based geospatial catalogues using natural language queries. The system converts user intent into structured API calls, enabling efficient access to...

https://arxiv.org/abs/2606.15077

1 **CONF**

Cognitive Debt: AI as Intellectual Leverage and the Dynamics of Systemic Fragility

ArXiv CS.AI **50%**

arXiv:2606.15078v1 Announce Type: new Abstract: We develop a formal theory of cognitive debt: the stock of unverified reasoning obligations that accumulates when individuals use AI as a substitute rather than a complement for first-principles cognition. The model features two...

https://arxiv.org/abs/2606.15078

2 **CONF**

VGPT-RSI for RH-Adjacent Formal Progress: Boundary Certificates, Verified Finite Lagarias Inequalities, and Explicit Failure Localization

ArXiv CS.AI **50%**

arXiv:2606.15096v1 Announce Type: new Abstract: The Riemann Hypothesis remains one of the central unsolved problems in mathematics. Rather than claiming proof, we investigate whether a verifiable AI-assisted reasoning system can produce reliable, formally checked partial...

https://arxiv.org/abs/2606.15096

3 **CONF**

CONCORD: Asynchronous Sparse Aggregation for Device-Cloud RAG under Document Isolation

ArXiv CS.AI **50%**

arXiv:2606.15179v1 Announce Type: new Abstract: Retrieval-augmented generation (RAG) has emerged as a pivotal technique for improving language models by incorporating external knowledge at inference time. As device-cloud collaborative inference makes it feasible to deploy small...

https://arxiv.org/abs/2606.15179

4 **CONF**

Scalable Circuit Learning for Interpreting Large Language Models

ArXiv CS.AI **50%**

arXiv:2606.16939v1 Announce Type: cross Abstract: A prominent research direction in mechanistic interpretability is learning sparse circuits over LLM components to reveal how they jointly produce model behavior. However, raw neurons are polysemantic, making learned circuits hard...

https://arxiv.org/abs/2606.16939

5 **CONF**

RollArt: Disaggregated Multi-Task Agentic RL Training at Scale

ArXiv CS.AI **50%**

arXiv:2512.22560v2 Announce Type: replace-cross Abstract: Agentic Reinforcement Learning (RL) trains LLMs through multi-turn interactions with environments, producing workloads that mix compute-bound prefill, bandwidth-bound decoding, CPU-heavy environment execution, and bursty...

https://arxiv.org/abs/2512.22560

6 **CONF**

Z-Plane Neural Networks: Bounded Geometric Activation Replaces ReLU and LayerNorm

ArXiv CS.AI **50%**

arXiv:2606.15669v1 Announce Type: cross Abstract: Modern deep neural networks rely on Euclidean scalar activations (e.g., ReLU) and global normalization techniques (e.g., LayerNorm) to prevent gradient instability in deep architectures. However, these mechanisms inherently cause...

https://arxiv.org/abs/2606.15669

7 **CONF**

Cascaded Sparse Autoencoders Learn Multi-Level Visual Concepts in Multimodal LLMs

ArXiv CS.AI **50%**

arXiv:2606.16193v1 Announce Type: cross Abstract: Multimodal Large Language Models (MLLMs) have demonstrated strong performance on vision-language tasks, yet their internal visual representations remain difficult to interpret. Sparse Autoencoders (SAEs) provide a scalable way to...

https://arxiv.org/abs/2606.16193

8 **CONF**

MatchLM2Lite: A Scalable MLLM-to-Lite Framework for Reproduced Content Identification

ArXiv CS.AI **50%**

arXiv:2606.14786v1 Announce Type: cross Abstract: Content moderation is critical for online video platforms to ensure content safety, protect creators, and sustain positive user experiences. Beyond filtering harmful content, platforms must guarantee content authenticity at scale...

https://arxiv.org/abs/2606.14786

9 **CONF**

APEX: Adaptive Principle EXtraction A Three-Layer Self-Evolution Framework for Production AI Agents

ArXiv CS.AI **50%**

arXiv:2606.15363v1 Announce Type: new Abstract: Self-improvement in AI agents has emerged as a key research frontier: systems that modify their own prompts, workflows, and decision rules based on accumulated operational experience. The state-of-the-art Self-Harness framework [1]...

https://arxiv.org/abs/2606.15363

0 **CONF**

Reward Hacking in Language Model Agents: Revisiting AI Safety Gridworlds

ArXiv CS.AI **50%**

arXiv:2606.15385v1 Announce Type: new Abstract: Reward hacking, where AI systems exploit misspecified objectives to achieve high reward without satisfying intended goals, remains a central challenge in AI safety. Yet most known instances have been discovered post hoc in frontier...

https://arxiv.org/abs/2606.15385

1 **CONF**

Graphical-Probabilistic Modeling of Generative Flows in LLM-Native Software Systems

ArXiv CS.AI **50%**

arXiv:2606.15943v1 Announce Type: cross Abstract: Engineering LLM-native software remains a challenging and immature field. Current practice is largely exploratory, relying on experimentation and heuristic techniques such as prompting and context engineering. These, however, are...

https://arxiv.org/abs/2606.15943

2 **CONF**

Who Drifted: the System or the Judge? Anytime-Valid Attribution in LLM Evaluation Pipelines

ArXiv CS.AI **50%**

arXiv:2606.15474v1 Announce Type: new Abstract: Continuous evaluation of LLM products relies on a strong LLM judge treated as ground truth: a cheap monitor scores every interaction and a team is paged when the score drifts down. But the judge is itself a model behind an API, and...

https://arxiv.org/abs/2606.15474

3 **CONF**

Towards End-to-End Automation of AI Research

ArXiv CS.AI **50%**

arXiv:2606.15497v1 Announce Type: new Abstract: The automation of science is a long-standing ambition in the field of AI. While the community has made significant progress in automating individual components of the scientific process, a system that autonomously navigates the...

https://arxiv.org/abs/2606.15497

4 **CONF**

Synthetic Counteradaptation: A Principle of Human-AI Co-evolution

ArXiv CS.AI **50%**

arXiv:2606.15503v1 Announce Type: new Abstract: In this paper, we introduce the concept of synthetic counteradaptation, a process where human and AI systems co-evolve by adapting to each other's strategies and behaviors. Synthetic counteradaptation occurs when AI systems develop...

https://arxiv.org/abs/2606.15503

5 **CONF**

Frame-Conditioned Moral Computation in LLaMA 3.1-8B-Instruct: A Mechanistic Interpretability Audit of Ethical Reasoning

ArXiv CS.AI **50%**

arXiv:2606.15507v1 Announce Type: new Abstract: Behavioral audits of Large Language Models on moral prompts measure what the model says, not the internal computation producing it. We use Transluce, an AI-driven mechanistic-interpretability platform, to examine LLaMA...

https://arxiv.org/abs/2606.15507

6 **CONF**

ROSA-RL: Uncertainty-Aware Roundabout Optimized Speed Advisory with Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2606.16558v1 Announce Type: new Abstract: Roundabouts challenge automated driving in mixed traffic, as heterogeneous and non-deterministic human behavior, unknown driving intentions, and high interaction complexity create uncertainty about whether the conflict zone will be...

https://arxiv.org/abs/2606.16558

7 CONF

Do we have the knowledge we need? Rethinking human-AI decision-making in corporations

ArXiv CS.AI 50%

arXiv:2606.15575v1 Announce Type: new Abstract: Organizational knowledge is fragmented across a variety of software systems, tacit expertise, and manual documents that have traditionally been designed for human consumption. As AI systems are increasingly deployed and granted...

https://arxiv.org/abs/2606.15575

8 CONF

Large Language Models as Optimizers: A Survey of Direct vs. Tool-Augmented Approaches and Their Performance Frontiers

ArXiv CS.AI 50%

arXiv:2606.15577v1 Announce Type: new Abstract: Large Language Models (LLMs) are increasingly involved in complex mathematical optimization, even if the pragmatic user who triggers them is unaware of it. After all, many real-world problems reduce to the search for better or the...

https://arxiv.org/abs/2606.15577

9 CONF

Your Agent Has a Genome: Sequence-Level Behavioral Analysis and Runtime Governance of LLM-Powered Autonomous Agents

ArXiv CS.AI 50%

arXiv:2606.15579v1 Announce Type: new Abstract: We propose Base Sequence Analysis, a framework that encodes the runtime behavior of LLM-powered autonomous agents into compact symbolic sequences using a four-letter alphabet: X (Explore), E (Execute), P (Plan), and V (Verify)....

https://arxiv.org/abs/2606.15579

0 CONF

Agentic Retrieval and Reinforcement Learned Equation Chains: A Controlled Generation Framework for Complex and Novel Physics Word Problems

ArXiv CS.AI 50%

arXiv:2606.15591v1 Announce Type: new Abstract: Generating high-quality Physics Word Problems (PWPs) that are novel, complex, and solvable remains a challenging and underexplored problem in educational content generation. Existing approaches, many adapted from Math Word Problem...

https://arxiv.org/abs/2606.15591

1 CONF

Integrating Reasoning and Generalization in Text-to-SQL via Self-Enhanced Fine-Tuning

ArXiv CS.AI 50%

arXiv:2606.15598v1 Announce Type: new Abstract: Text-to-SQL aims to translate natural language questions into executable SQL queries over structured databases, enabling non-expert users to access data intuitively. While recent advances in large language models (LLMs) have shown...

https://arxiv.org/abs/2606.15598

2 CONF

NeuroSymbolic AI for Legal AI-TRISM: Trustworthy, Reliable, Interpretable, Safe Models

ArXiv CS.AI 50%

arXiv:2606.15646v1 Announce Type: new Abstract: Large Language Models (LLMs) have transformed natural language processing, but their lack of interpretable reasoning and tendency to hallucinate pose significant challenges for legal applications. While LLMs show promise for legal...

https://arxiv.org/abs/2606.15646

3 **CONF**

Towards Next-Generation Healthcare: A Survey of Medical Embodied AI for Perception, Decision-Making, and Action

ArXiv CS.AI 50%

arXiv:2606.15647v1 Announce Type: new Abstract: Foundation models have demonstrated impressive performance in enhancing healthcare efficiency across a wide range of medical applications. Nevertheless, their limited ability to perceive, understand, and interact with the physical...

https://arxiv.org/abs/2606.15647

4 **CONF**

Advanced Machine Learning and Deep Learning Techniques for Enhanced Cattle Identification and Detection: A Comprehensive Review

ArXiv CS.AI 50%

arXiv:2606.15655v1 Announce Type: new Abstract: The need for effective cattle identification technology is now more acutely felt than ever in maintaining biosecurity, food safety, and supply chain efficacy in livestock management. This paper presents a systematic review of...

https://arxiv.org/abs/2606.15655

5 **CONF**

Overcoming the Impedance Mismatch: A Theoretical Roadmap for Fusing Foundation Models and Knowledge Graphs

ArXiv CS.AI 50%

arXiv:2606.15656v1 Announce Type: new Abstract: Modern artificial intelligence remains fundamentally divided between the continuous, probabilistic spaces of Foundation Models and the discrete, deterministic structures of Knowledge Graphs. While Retrieval-Augmented Generation...

https://arxiv.org/abs/2606.15656

6 **CONF**

Do LLMs Reliably Identify Correct Information Units in Aphasic Discourse?

ArXiv CS.AI 50%

arXiv:2606.15696v1 Announce Type: new Abstract: Correct Information Units (CIUs) are central to discourse assessment in aphasia because they quantify communicative informativeness rather than linguistic form alone. However, CIU scoring is time intensive and requires trained...

https://arxiv.org/abs/2606.15696

7 **CONF**

Artificial Intelligence Index Report 2026

ArXiv CS.AI 50%

arXiv:2606.15708v1 Announce Type: new Abstract: Welcome to the ninth edition of the AI Index report. As AI continues to advance rapidly, the question becomes whether the systems built around it can keep up. Governance frameworks, evaluation methods, education systems, and the...

https://arxiv.org/abs/2606.15708

8 **CONF**

AI-Driven Framework for Adaptive Water Network Management with Proof-of-Concept Implementation: Addressing Non-Revenue Water in Jordan

ArXiv CS.AI 50%

arXiv:2606.15709v1 Announce Type: new Abstract: Jordan faces severe water scarcity with 50\% of water produced is lost to leakage, theft and metering issues also known as non-revenue water (NRW). Traditional reactive approaches have proven insufficient for sustained NRW...

https://arxiv.org/abs/2606.15709

9 CONF

RoboPIN: Grounded Embodied Reasoning via Pinned Chain-of-Thought

ArXiv CS.AI 50%

arXiv:2606.15753v1 Announce Type: new Abstract: Embodied reasoning requires models to perceive task-relevant objects and spaces in physical environments and maintain consistent visual grounding throughout multi-step reasoning. However, current vision-language models rely on...

https://arxiv.org/abs/2606.15753

0 CONF

Rethinking Scaffolding in LLM Tutors: The Interactional Mismatch Between Benchmarks and Real-World Deployments

ArXiv CS.AI 50%

arXiv:2606.15766v1 Announce Type: new Abstract: A central pedagogical value evaluated in AI tutor benchmarks is scaffolding: guiding students through graduated steps toward a solution. Alignment and evaluation methods for embedding scaffolding behaviour into chatbots, however,...

https://arxiv.org/abs/2606.15766

1 CONF

Lost at the End: Primacy Bias in Multimodal Retrieval-Augmented Question Answering

ArXiv CS.AI 50%

arXiv:2606.16494v1 Announce Type: cross Abstract: Knowledge-based visual question answering (KB-VQA) lets vision-language systems answer questions that exceed their parametric knowledge by conditioning a reader on passages retrieved from a Wikipedia-scale knowledge base. In...

https://arxiv.org/abs/2606.16494

2 CONF

An Integrated System for Real-Time Student Assessment and Career Guidance Using Neural Networks in Computing Disciplines

ArXiv CS.AI 50%

arXiv:2606.15831v1 Announce Type: new Abstract: Many undergraduate students in Computer Science (CS) and Software Engineering (SWE) struggle to identify suitable career paths, particularly when their academic performance, abilities, and interests do not fully align. To address...

https://arxiv.org/abs/2606.15831

3 CONF

Architectural Wisdom: A Framework for Governing Optimization in AI Systems

ArXiv CS.AI 50%

arXiv:2606.16319v1 Announce Type: new Abstract: Modern AI systems exhibit structural failures that capability scaling alone does not reliably fix: they optimize under-specified objectives with no architectural mechanism to question whether the objective should be optimized at...

https://arxiv.org/abs/2606.16319

4 CONF

Heteroskedastic Signals in Budgeted LLM Verification: Structural Heterogeneity Limits Optimization Gains

ArXiv CS.AI 50%

arXiv:2606.15841v1 Announce Type: new Abstract: Large language model (LLM) systems increasingly use uncertainty signals to allocate limited computation across verification, test-time scaling, tool execution, and other selective-compute decisions. Such policies rely on a...

https://arxiv.org/abs/2606.15841

5 **CONF**

Agentic Framework for Deep Learning workload migration via In-Context Learning

ArXiv CS.AI **50%**

arXiv:2606.15994v1 Announce Type: new Abstract: Translating deep learning models from PyTorch's flexible, object-oriented design to JAX's functional, stateless setup is usually a manual and error-prone task. Automated migration is challenging because Large Language Models (LLMs)...

https://arxiv.org/abs/2606.15994

6 **CONF**

LLM-as-Code Agentic Programming for Agent Harness

ArXiv CS.AI **50%**

arXiv:2606.15874v1 Announce Type: new Abstract: Every major LLM agent framework gives the LLM the role of orchestrator; the model decides what to do next, when to call tools, and when to stop. We argue that token explosion, control-flow hallucination, and unreliable completion...

https://arxiv.org/abs/2606.15874

7 **CONF**

EffGen: Enabling Small Language Models as Capable Autonomous Agents

ArXiv CS.AI **50%**

arXiv:2602.00887v2 Announce Type: replace-cross Abstract: Most existing language model agentic systems today are built and optimized for large language models (e.g., GPT, Claude, Gemini) via API calls; while powerful, this approach faces several limitations including high token...

https://arxiv.org/abs/2602.00887

8 **CONF**

Implicit Reasoning for Large Language Model-based Generative Recommendation

ArXiv CS.AI **50%**

arXiv:2606.14142v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) are increasingly adopted as backbones for Generative Recommendation (GR), promising access to pretrained world knowledge. Yet reliably invoking this knowledge for GR remains poorly understood....

https://arxiv.org/abs/2606.14142

9 **CONF**

Know Your Limits : On the Faithfulness of LLMs as Solvers and Autoformalizers in Legal Reasoning

ArXiv CS.AI **50%**

arXiv:2606.16118v1 Announce Type: new Abstract: Large Language Models (LLMs) achieve strong performance on reasoning tasks, but whether this reflects faithful logical inference or heuristic approximation remains unclear. We study this question in legal entailment by comparing...

https://arxiv.org/abs/2606.16118

0 **CONF**

Phishing Email Detection Using Large Language Models

ArXiv CS.AI **50%**

arXiv:2512.10104v2 Announce Type: cross Abstract: Email phishing is one of the most prevalent and globally consequential vectors of cyber intrusion. As systems increasingly deploy Large Language Models (LLMs) applications, these systems face evolving phishing email threats that...

https://arxiv.org/abs/2512.10104

1 CONF

Medical Heuristic Learning: An LLM-Driven Framework for Interpretable and Auditable Clinical Decision Rules

ArXiv CS.AI 50%

arXiv:2606.16337v1 Announce Type: new Abstract: Predictive modeling for clinical tabular data is central to clinical decision support and therefore requires not only strong predictive performance but also transparent decision logic. Although deep learning and tree-based ensemble...

https://arxiv.org/abs/2606.16337

2 CONF

The Quality-Utility Paradox: Why High-Reward Data Impairs Small Model Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2606.16152v1 Announce Type: new Abstract: Knowledge distillation from powerful reasoning models is widely used to improve Small Language Models (SLMs) on mathematical reasoning, often assuming that traces with higher reward model scores provide more useful supervision. We...

https://arxiv.org/abs/2606.16152

3 CONF

AI Pluralism and the Worlds It Misses

ArXiv CS.AI 50%

arXiv:2606.16167v1 Announce Type: new Abstract: AI pluralism is often framed as a problem of representing diverse values, preferences, users, or outputs. This paper argues that this framing is incomplete because AI systems also impose ontologies: they define what counts as an...

https://arxiv.org/abs/2606.16167

4 CONF

Retrieve, Don't Retrain: Extending Vision Language Action Models to New Tasks at Test Time

ArXiv CS.AI 50%

arXiv:2606.15631v1 Announce Type: cross Abstract: Extending a vision-language-action (VLA) policy to a new task typically requires task-specific teleoperated demonstrations and per-task fine-tuning, making adaptation costly in both data collection and compute. In this paper, we...

https://arxiv.org/abs/2606.15631

5 CONF

Measuring Whether LLM Tutors Teach or Solve: A Diagnostic for Educational Impact

ArXiv CS.AI 50%

arXiv:2606.16206v1 Announce Type: new Abstract: Large language models are increasingly proposed as educational tutors, yet stronger task-solving ability does not necessarily imply stronger learning support. Motivated by recent calls to measure the social impact of NLP systems in...

https://arxiv.org/abs/2606.16206

6 CONF

SpecAlign: Efficient Specification-Grounded Alignment of Large Language Models via Synthetic Data

ArXiv CS.AI 50%

arXiv:2606.16276v1 Announce Type: new Abstract: As large language models (LLMs) are increasingly deployed in real-world applications, alignment is no longer governed by a single universal notion of safety or helpfulness, but instead by provider- or application-specific model...

https://arxiv.org/abs/2606.16276

7 CONF

AdaSTORM: Scaling LLM Reasoning on Dynamic Graphs via Adaptive Spatio-Temporal Multi-Agent Collaboration

ArXiv CS.AI 50%

arXiv:2606.16328v1 Announce Type: new Abstract: Large Language Models (LLMs) demonstrate remarkable potential in dynamic graph reasoning, but suffer from a scaling bottleneck: current models can only handle graphs with tens of nodes, constrained by exponential reasoning overhead...

https://arxiv.org/abs/2606.16328

8 CONF

LLM Judges Have Dark Current: A Psychometric Datasheet for LLM-as-a-Judge Evaluation

ArXiv CS.AI 50%

arXiv:2606.15610v1 Announce Type: cross Abstract: LLM-as-a-judge systems are now routinely used for open-ended model evaluation, where human preference annotation is costly, slow, and difficult to reproduce. Yet these judges are often reported as scalar accuracy, win-rate, or...

https://arxiv.org/abs/2606.15610

9 CONF

Decision-Aware Memory Cards: Counterfactual-Inspired Context Selection and Compression for Tool-Using LLM Agents

ArXiv CS.AI 50%

arXiv:2606.08151v2 Announce Type: replace Abstract: Modern large language model (LLM) agents do not simply need longer contexts; they need decision-relevant evidence at the moment of action. We study decision-aware context selection: ranking retrieved files, tests, traces,...

https://arxiv.org/abs/2606.08151

0 CONF

When Agent Automation Becomes Profitable: Quantifying and Insuring Autonomous AI Risk through Trace-Economic Underwriting

ArXiv CS.AI 50%

arXiv:2606.16465v1 Announce Type: new Abstract: AI agents can now take irreversible actions in operational systems, but agent-caused losses are still not clearly assigned, priced, or transferred. Providers often disclaim consequential damages, users are left with uncompensated...

https://arxiv.org/abs/2606.16465

1 CONF

Tensor-Coord: Algebraic Decomposition of Joint Plan Tensors for Conflict-Free Multi-Agent LLM Planning

ArXiv CS.AI 50%

arXiv:2606.16478v1 Announce Type: new Abstract: Large language models (LLMs) remain limited in multi-agent planning because independently generated plans can create coordination failures such as spatial collisions, resource contention, and temporal deadlocks. We introduce...

https://arxiv.org/abs/2606.16478

2 CONF

Steering Emotional Dynamics for Art Therapy: Controllable Narrative Script Generation through Hierarchically Guided LLM Agents

ArXiv CS.AI 50%

arXiv:2606.16481v1 Announce Type: new Abstract: Art therapy plays a vital role in emotional healing, in which narrative creation acts as the primary vehicle for emotional expression. Given the inherently dynamic nature of emotions during healing, narratives with finely...

https://arxiv.org/abs/2606.16481

3 CONF

Kairos: A Native World Model Stack for Physical AI

ArXiv CS.AI 50%

arXiv:2606.16533v1 Announce Type: new Abstract: World models are transitioning from passive visual generators to foundational, operational infrastructure for Physical AI: they must natively acquire world knowledge from heterogeneous experience, maintain persistent states over...

https://arxiv.org/abs/2606.16533

4 CONF

The Faithfulness Gap: Certifying Semantic Equivalence Between Natural-Language and Formal Mathematical Statements

ArXiv CS.AI 50%

arXiv:2606.16541v1 Announce Type: new Abstract: Autoformalization, translating natural-language mathematics into formal proof assistants, is bottlenecked not by translation fluency but by \emph{faithfulness}: a formal statement can typecheck and be provable, yet still encode a...

https://arxiv.org/abs/2606.16541

5 CONF

ARB4WM: An Adversarial Robustness Benchmark for World Models in Continuous Control

ArXiv CS.AI 50%

arXiv:2606.16605v1 Announce Type: new Abstract: World models are widely used in robotic and agentic engineering control systems due to their ability to learn latent dynamics for planning and decision-making. As these systems are increasingly deployed in safety-critical settings,...

https://arxiv.org/abs/2606.16605

6 CONF

MR-GVNO: A Geometry-Aware Variational Physics-Informed Neural Operator for Mindlin-Reissner Plates on Irregular Domains

ArXiv CS.AI 50%

arXiv:2606.16624v1 Announce Type: new Abstract: Plate and shell structures are widely used in engineering, making rapid response prediction under varying geometries, materials, and loads highly desirable. However, conventional finite element methods require repeated modeling and...

https://arxiv.org/abs/2606.16624

7 CONF

The Integrator Advantage: Controlled Agentic AI for Small and Medium-Sized Companies

ArXiv CS.AI 50%

arXiv:2606.16649v1 Announce Type: new Abstract: Agentic AI marks a new phase of enterprise automation. Unlike traditional automation or conversational AI, agentic systems can interpret goals, plan multi step tasks, access tools, interact with enterprise systems, and execute...

https://arxiv.org/abs/2606.16649

8 CONF

Is Your Agent Playing Dead? Deployed LLM Agents Exhibit Constraint-Evasive Fabrication and Thanatosis

ArXiv CS.AI 50%

arXiv:2606.14831v1 Announce Type: cross Abstract: This paper presents and characterizes a spectrum of previously unreported behaviours we term Constraint-Evasive Fabrication (CEF): when an LLM agent operates under irreconcilable constraints (where no response can simultaneously...

https://arxiv.org/abs/2606.14831

9 **CONF**

A First-Principles Derivation of LLM Policy Optimization: From Expected Reward to GRPO and Its Structural Extensions

ArXiv CS.AI **50%**

arXiv:2606.16733v1 Announce Type: new Abstract: Policy gradient algorithms for language models optimize the same objective $J(\theta) = \mathbb{E}*{\tau \sim p*\theta(\tau)}[R(\tau)]$, which has exactly two factors: the trajectory probability $p_\theta(\tau)$ and the reward...

https://arxiv.org/abs/2606.16733

0 **CONF**

Metacognitive Myopia in Large Language Models

ArXiv CS.AI **50%**

arXiv:2408.05568v2 Announce Type: replace Abstract: Large Language Models (LLMs) exhibit potentially harmful biases that reinforce culturally embedded stereotypes, influence moral judgments, or amplify positive evaluations of majority groups. We propose metacognitive myopia as a...

https://arxiv.org/abs/2408.05568

1 **CONF**

Scaling LLM Reasoning from Minimal Labels: A Semi-Supervised Framework with a Lightweight Verifier

ArXiv CS.AI **50%**

arXiv:2606.16811v1 Announce Type: new Abstract: For the development of Large language models (LLMs), recent approaches to generating pseudo intermediate reasoning have shown remarkable progress. But they typically rely on large numbers of correctly annotated answers to assess...

https://arxiv.org/abs/2606.16811

2 **CONF**

DifFRACT: Diffusion Feature Reconstruction and Attribution for Circuit Tracing

ArXiv CS.AI **50%**

arXiv:2606.15796v1 Announce Type: cross Abstract: Mechanistic interpretability seeks to explain neural network behavior by decomposing model computations into interpretable features and circuits. While transcoder-based circuit tracing has recently enabled detailed causal...

https://arxiv.org/abs/2606.15796

3 **CONF**

The embrace of open science: An analysis of a decade of AI research and 56 800 conference papers

ArXiv CS.AI **50%**

arXiv:2606.16974v1 Announce Type: new Abstract: The reproducibility crisis has directed the AI research community toward improving documentation practices. Several studies have identified methodological issues, and in response, the most impactful venues in the field have...

https://arxiv.org/abs/2606.16974

4 **CONF**

Bayesian Inference and Decision Audits for Public Archives of Frontier AI Evaluations

ArXiv CS.AI **50%**

arXiv:2606.17005v1 Announce Type: new Abstract: Public AI evaluations are often read as terminal leaderboards, yet the underlying evidence is a selective time series shaped by reporting rules, benchmark revisions, and missingness. Repeated public archives for LiveBench and Open...

https://arxiv.org/abs/2606.17005

5 CONF

Limited Marginal Benefit of Reasoning-Heavy LLM Deployment in ESG Narrative Scoring: A 4-Model Consensus Study on Japanese Listed Firms

ArXiv CS.AI 50%

arXiv:2606.13693v1 Announce Type: cross Abstract: Automated scoring of ESG narrative disclosures with large language models (LLMs) is gaining traction, yet whether reasoning-heavy frontier models add value commensurate with their cost remains empirically unsettled. We evaluate...

https://arxiv.org/abs/2606.13693

6 CONF

Green AI Carbon Optimizer: Carbon-Efficient Training Location Recommendation and Global AI Energy Demand Forecasting

ArXiv CS.AI 50%

arXiv:2606.14707v1 Announce Type: cross Abstract: AI training and deployment consume substantial electricity, but carbon outcomes remain weakly integrated into routine model development decisions. This paper presents Green AI Carbon Optimizer with two primary contributions: (i)...

https://arxiv.org/abs/2606.14707

7 CONF

MiroBench: Benchmarking Realism in Agentic Simulation of Real-world Discussions

ArXiv CS.AI 50%

arXiv:2606.14715v1 Announce Type: cross Abstract: LLM agents are increasingly used to simulate real world interactions, but it remains unclear whether simulated behaviors preserve the content patterns and interaction dynamics of real human behaviors. Existing evaluations remain...

https://arxiv.org/abs/2606.14715

8 CONF

Gender Differences in AI Literacy Workshop Outcomes and Deepfake Engagement

ArXiv CS.AI 50%

arXiv:2606.14718v1 Announce Type: cross Abstract: As Artificial Intelligence (AI) literacy initiatives expand in K-12 settings, understanding how gender shapes student baseline perceptions, tool-use, and responsiveness to interventions is essential for equitable curriculum...

https://arxiv.org/abs/2606.14718

9 CONF

Communication-Efficient Verifiable Attention for LLM Inference

ArXiv CS.AI 50%

arXiv:2606.16352v1 Announce Type: cross Abstract: Computation integrity of remote large language model (LLM) serving can be questionable. For conventional deep neural networks (DNNs), the existing TEE-shielded DNN partitioning (TSDP) approach uses Trusted Execution Environment...

https://arxiv.org/abs/2606.16352

00 CONF

Steady-Forcing: Balancing Spatial Persistence and Motion Continuity in Long-Horizon Nature Video Diffusion

ArXiv CS.AI 50%

arXiv:2606.14732v1 Announce Type: cross Abstract: Autoregressive video diffusion models enable streaming generation but often degrade over long rollouts: static scene layouts drift, while mechanisms that improve spatial stability tend to suppress motion, causing natural flows...

https://arxiv.org/abs/2606.14732

01 CONF

Are Neuro-Inspired Multi-Modal Vision-Language Models Resilient to Membership Inference Privacy Leakage?

ArXiv CS.AI 50%

arXiv:2511.20710v2 Announce Type: replace-cross Abstract: In the age of agentic AI, the growing deployment of multi-modal models (MMs) has introduced new attack vectors that can leak sensitive training data in MMs, causing privacy leakage. This paper investigates a black-box...

https://arxiv.org/abs/2511.20710

02 CONF

Unified Multimodal Model for Brain MRI Imputation and Understanding

ArXiv CS.AI 50%

arXiv:2606.16484v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) hold great potential for medicine, as they inherit knowledge from LLM and allow multiple data modalities to be integrated, analysed and interpreted in natural language. However, the field...

https://arxiv.org/abs/2606.16484

03 CONF

Beyond Self-Attention: Sub-Quadratic Vision Transformers for Fast Image Captioning

ArXiv CS.AI 50%

arXiv:2606.14753v1 Announce Type: cross Abstract: Image captioning is a challenging and significant task that aims to generate coherent and semantically meaningful textual descriptions for given images. To accomplish this task, it requires a deep understanding of visual content...

https://arxiv.org/abs/2606.14753

04 CONF

GeoRoPE: Ground-Aware Rotary Adaptation for Remote Sensing Foundation Models

ArXiv CS.AI 50%

arXiv:2606.14760v1 Announce Type: cross Abstract: Remote-sensing foundation models (RSFMs) benefit from pretraining on imagery from multiple sensors and ground sampling distances (GSDs), but such exposure alone does not resolve scale mismatch during downstream adaptation. A...

https://arxiv.org/abs/2606.14760

05 CONF

Mosaic: Data-Free Knowledge Distillation via Mixture-of-Experts for Heterogeneous Distributed Environments

ArXiv CS.AI 50%

arXiv:2505.19699v2 Announce Type: replace-cross Abstract: Federated Learning (FL) is a decentralized machine learning paradigm that enables clients to collaboratively train models while preserving data privacy. However, the coexistence of model and data heterogeneity gives rise...

https://arxiv.org/abs/2505.19699

06 CONF

Agentomics: Economic Foundations for the Valuation, Attribution, and Pricing of AI Agents in Human-AI Workflows

ArXiv CS.AI 50%

arXiv:2606.14769v1 Announce Type: cross Abstract: Agentic AI systems are increasingly being deployed as productive resources in organizational workflows, yet existing evaluation methods primarily measure isolated technical performance rather than economic contribution. This...

https://arxiv.org/abs/2606.14769

07 **CONF**

Double-Helix Vision (DH-V2): A Geometry-Based Visual Sampler for Bandwidth-Constrained Perception

ArXiv CS.AI 50%

arXiv:2606.14773v1 Announce Type: cross Abstract: We present Double-Helix Vision (DH), a geometry-based visual sampler that compresses 2D images into compact 1D signals using paired golden-ratio-inspired spiral trajectories. Rather than processing every pixel uniformly, DH...

https://arxiv.org/abs/2606.14773

08 **CONF**

Unifying Acoustic Features and Text with Multimodal LLMs for Neurodegenerative Screening

ArXiv CS.AI 50%

arXiv:2606.14788v1 Announce Type: cross Abstract: Voice-based screening offers a scalable and non-invasive way to assess neurodegenerative diseases such as Alzheimer's disease (AD) and Parkinson's disease (PD), but their staging remains challenging due to the difficulty of...

https://arxiv.org/abs/2606.14788

09 **CONF**

Fast When, Careful Who: Dual-Process Multiparty Turn-Taking with Diffusion Augmentation

ArXiv CS.AI 50%

arXiv:2606.16568v1 Announce Type: cross Abstract: Reliable turn-taking is essential for spoken dialogue systems. However, most existing methods are designed for two-speaker interaction and struggle with realistic multiparty audio containing overlap and rapid speaker changes. We...

https://arxiv.org/abs/2606.16568

10 **CONF**

A Security Analysis of Long-Horizon Agentic AI Systems: Threats, Evaluation, and Framework Development

ArXiv CS.AI 50%

arXiv:2606.14816v1 Announce Type: cross Abstract: This paper presents a structured analysis of security challenges in long-horizon agentic AI systems. The study reviews existing threats, evaluation approaches, attack propagation mechanisms, and security frameworks. A taxonomy of...

https://arxiv.org/abs/2606.14816

11 **CONF**

PVminerLLM2: Improving Structured Extraction of Patient Voice via Preference Optimization

ArXiv CS.AI 50%

arXiv:2606.16074v1 Announce Type: cross Abstract: Motivation: Patient-generated text contains critical information on patients' lived experiences, social context, and care engagement, but remains largely unstructured, limiting its use in patient-centered outcomes research. Prior...

https://arxiv.org/abs/2606.16074

12 **CONF**

Spectro-Temporal Interference Confounds Phase Encoding in Spatial Audio Foundation Models

ArXiv CS.AI 50%

arXiv:2606.14820v1 Announce Type: cross Abstract: Recent spatial self supervised audio models achieve high performance on localization tasks, raising questions about their encoding of microsecond interaural phase fine structures. We propose a psychoacoustic benchmark based on...

https://arxiv.org/abs/2606.14820

13 CONF

Quantum Machine Learning for Industrial Applications

ArXiv CS.AI 50%

arXiv:2606.14822v1 Announce Type: cross Abstract: Recent advances in Machine Learning have transformed numerous industrial sectors, yet classical paradigms face fundamental limitations: rapidly growing data volumes, rising computational costs, significant energy consumption, and...

https://arxiv.org/abs/2606.14822

14 CONF

Human genetic evidence is associated with drug approval across therapeutic areas: an observational analysis of 26,278 target-disease pairs with temporal validation and feature ablation

ArXiv CS.AI 50%

arXiv:2606.14823v1 Announce Type: cross Abstract: Genetic evidence is enriched among approved drug targets: in an observational analysis of 26,278 target-disease pairs from Open Targets and ChEMBL, targets with any genetic association had a 3.25-fold higher approval rate than...

https://arxiv.org/abs/2606.14823

15 CONF

Leptomeningeal Collateral Detection on DSA via Vessel-Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2606.14828v1 Announce Type: cross Abstract: Leptomeningeal collaterals (LMCs) are an important prognostic factor in acute ischemic stroke. Existing automated methods rely on CT angiography (CTA), but individual LMCs are often too small to be resolved on CTA, limiting these...

https://arxiv.org/abs/2606.14828

16 CONF

An Ensemble Deep Learning Approach for Reliable and Scalable Lemon Leaf Disease Classification

ArXiv CS.AI 50%

arXiv:2606.14871v1 Announce Type: cross Abstract: Early detection of plant diseases is crucial to plants and for the farmers. Plant diseases reduce fruit yield and quality, and plants are more susceptible to other stresses when they are infected. The lemon leaf disease dataset...

https://arxiv.org/abs/2606.14871

17 CONF

Policy Regret for Embedding Model Routing: Contextual Bandits with Low-Rank Experts

ArXiv CS.AI 50%

arXiv:2606.14929v1 Announce Type: cross Abstract: Modern recommendation systems increasingly rely on dynamically routing diverse queries to multiple embedding models. Despite its practical significance, this problem remains poorly understood under realistic conditions like...

https://arxiv.org/abs/2606.14929

18 CONF

Beyond Correctness: Enhancing Architectural Reasoning in Code LLMs via Scalable Labeling with Agentic Judgment

ArXiv CS.AI 50%

arXiv:2606.14948v1 Announce Type: cross Abstract: LLMs have substantially improved software engineering yet real-world development requires architectural understanding. Such understanding is prohibitively expensive to label manually and impossible to verify through tests alone....

https://arxiv.org/abs/2606.14948

19 CONF

Multi-Modal Attention for Automated Disaster Damage Assessment Using Remote Sensing Imagery and Deep Learning

ArXiv CS.AI 50%

arXiv:2606.14963v1 Announce Type: cross Abstract: Timely and accurate disaster damage assessment is crucial for effective emergency response, resource allocation, and recovery. Traditional methods, which often rely on manual inspections or sparse data, are typically slow and...

https://arxiv.org/abs/2606.14963

20 CONF

Inference-time Policy Steering via Vision and Touch

ArXiv CS.AI 50%

arXiv:2606.14981v1 Announce Type: cross Abstract: Inference-time steering adapts pre-trained generative robot policies during deployment by verifying candidate actions before execution. While prior methods typically perform this verification only with visual observations, vision...

https://arxiv.org/abs/2606.14981

21 CONF

Nemotron 3 Ultra: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Model for Agentic Reasoning

ArXiv CS.AI 50%

arXiv:2606.15007v1 Announce Type: cross Abstract: We introduce Nemotron 3 Ultra, a 550 billion total and 55 billion active parameter Mixture-of-Experts Hybrid Mamba-Attention language model. We pre-trained Nemotron 3 Ultra on 20 trillion text tokens, then extended the context...

https://arxiv.org/abs/2606.15007

22 CONF

Resilient Consensus in Agentic AI

ArXiv CS.AI 50%

arXiv:2606.15024v1 Announce Type: cross Abstract: Large language model (LLM) agents are increasingly deployed in multi-agent systems where they must coordinate and agree on shared decisions. We ask whether classical resilient consensus theory, developed for deterministic agents,...

https://arxiv.org/abs/2606.15024

23 CONF

PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation

ArXiv CS.AI 50%

arXiv:2606.15052v1 Announce Type: cross Abstract: Traditional design of analog circuits heavily relies on manual interventions across topology, sizing, and layout, with prior automation addressing stages in isolation. In this work, we propose PANDA, an LLM-enhanced framework...

https://arxiv.org/abs/2606.15052

24 CONF

Bridging Geographic Bias in Urban Streetscape Inference via Lifelong Learning with Visual-Semantic Pivoting

ArXiv CS.AI 50%

arXiv:2606.15055v1 Announce Type: cross Abstract: Visual perception of urban streetscapes underpins evidence-based decisions in landscape planning, public health, and place-making. Yet models trained on a few well-photographed metropolises systematically misjudge...

https://arxiv.org/abs/2606.15055

25 CONF

AutoDojo: Adaptive Attacks Expose Superficial Defenses and User-Underspecification Limits in LLM Agents

ArXiv CS.AI 50%

arXiv:2606.15057v1 Announce Type: cross Abstract: Indirect prompt injection (IPI) is a major security threat to LLM-powered agents. Thus, a growing body of work have proposed a variety of defensive approaches against IPI. These can be grouped into three broad categories: 1)...

https://arxiv.org/abs/2606.15057

26 CONF

Mojo: A Promising Tool for Scalable Financial AI Efficiency

ArXiv CS.AI 50%

arXiv:2606.16059v1 Announce Type: cross Abstract: For thirty years, quantitative finance has paid a costly two-language tax: models researched in Python are rewritten in C++ for production, often introducing numerical discrepancies. GPU-accelerated deep learning exacerbates this...

https://arxiv.org/abs/2606.16059

27 CONF

Sensory Restoration via Brain-Computer Interfaces: A Unified 2 x 2 Framework and Convergence Roadmap

ArXiv CS.AI 50%

arXiv:2606.15091v1 Announce Type: cross Abstract: Millions of individuals worldwide suffer from sensory and communication deficits caused by neurodegenerative diseases, stroke, or trauma. Brain-computer interfaces (BCIs) offer a promising avenue for sensory and motor...

https://arxiv.org/abs/2606.15091

28 CONF

Teacher-Student Structure for Domain Adaptation in Ensemble Audio-Visual Video Deepfake Detection

ArXiv CS.AI 50%

arXiv:2606.15117v1 Announce Type: cross Abstract: The rapid advancement of generative AI models is leading to more realistic deepfake media, encompassing the manipulation of audio, video, or both. This raises severe privacy and societal concerns. Numerous studies in this area...

https://arxiv.org/abs/2606.15117

29 CONF

EyeMVP: OCT-Informed Fundus Representation Learning via Paired CFP--OCT Pretraining

ArXiv CS.AI 50%

arXiv:2606.15129v1 Announce Type: cross Abstract: Color fundus photography (CFP) is the mainstay for large-scale retinal screening, yet its diagnostic capacity is constrained by the lack of depth-resolved structural information. Optical coherence tomography (OCT) provides...

https://arxiv.org/abs/2606.15129

30 CONF

Beyond Scalar Distances: Semantic Attribute Gradients from Frozen MLLMs for Visual Embeddings

ArXiv CS.AI 50%

arXiv:2606.15134v1 Announce Type: cross Abstract: Vision encoders for retrieval are typically trained with class-label supervision: each training pair reduces to a scalar that uniformly pushes the embedding apart or pulls it together, as if every visual attribute either differed...

https://arxiv.org/abs/2606.15134

31 CONF

MimicIK: Real-Time Generative Inverse Kinematics from Teleoperation with FK Consistency

ArXiv CS.AI 50%

arXiv:2606.15148v1 Announce Type: cross Abstract: Inverse kinematics (IK) remains a critical bottleneck for real-time robot manipulation. Classical numerical solvers achieve high geometric precision but often suffer from discontinuous branch switching and unstable behavior near...

https://arxiv.org/abs/2606.15148

32 CONF

Utility-Diversity Aware Online Batch Selection for LLM Supervised Fine-tuning

ArXiv CS.AI 50%

arXiv:2510.16882v4 Announce Type: replace-cross Abstract: Supervised fine-tuning (SFT) is a commonly used technique to adapt large language models (LLMs) to downstream tasks. In practice, SFT on a full dataset is computationally expensive and sometimes suffers from overfitting...

https://arxiv.org/abs/2510.16882

33 CONF

Controlled Dynamics Attractor Transformer

ArXiv CS.AI 50%

arXiv:2606.15207v1 Announce Type: cross Abstract: Transformer architectures have dramatically advanced representation learning and inference in deep models through self-attention mechanisms. In parallel,associative memory (AM) frameworks map representations onto energy...

https://arxiv.org/abs/2606.15207

34 CONF

Spokes: Optimizing for Diverse Pretraining Data Selection

ArXiv CS.AI 50%

arXiv:2606.15216v1 Announce Type: cross Abstract: Diversity plays a critical role in data selection, improving performance under fixed data budgets by reducing redundancy and repetition. However, optimizing for diversity is inherently challenging, as it is a set-level property...

https://arxiv.org/abs/2606.15216

35 CONF

Landmark-free Assessment of Lower-limb Alignment with Implicit Neural Shape Functions from Knee Radiographs

ArXiv CS.AI 50%

arXiv:2606.15250v1 Announce Type: cross Abstract: Radiographic assessment of lower-limb alignment (LLA) is important for predicting joint health and surgical outcomes in total knee arthroplasty. Traditional measurement methods are manual and time-consuming, while recent machine...

https://arxiv.org/abs/2606.15250

36 CONF

Trust-Region Diffusion Policies for Massively Parallel On-Policy RL

ArXiv CS.AI 50%

arXiv:2606.15260v1 Announce Type: cross Abstract: Reinforcement learning with massively parallel simulations has become a standard framework for developing robust, deployable policies; however, most existing approaches still rely on simple Gaussian policy parameterizations....

https://arxiv.org/abs/2606.15260

37 CONF

Guiding Federated Graph Recommendation with LLM-encoded knowledge

ArXiv CS.AI 50%

arXiv:2606.15277v1 Announce Type: cross Abstract: Graph-based recommender systems are highly effective at extracting collaborative signals from user--item interactions, and federated learning (FL) allows these models to be trained while preserving user privacy. However,...

https://arxiv.org/abs/2606.15277

38 CONF

Hybrid NARX-LLM for Greenland Iceberg Discharge: Prompt-Driven Residual Correction

ArXiv CS.AI 50%

arXiv:2606.15288v1 Announce Type: cross Abstract: Greenland iceberg discharge exhibits complex nonlinear dynamics with limited observability, challenging traditional predictive models. We present a Hybrid NARX-LLM framework that combines a nonlinear autoregressive model with...

https://arxiv.org/abs/2606.15288

39 CONF

Robust Dual-Signal Fusion: Hybrid Neuro-Symbolic Gating with Compressed Chain-of-Thought Refinement for Irony Detection in Social Media Texts

ArXiv CS.AI 50%

arXiv:2606.16845v1 Announce Type: cross Abstract: Large Language Models (LLMs) natively default to literal semantic interpretations, making zero-shot irony detection a persistent challenge. We introduce the Robust Dual-Signal (RDS) Fusion framework, a hybrid neuro-symbolic...

https://arxiv.org/abs/2606.16845

40 CONF

LearnOpt: Recovering the Latent Cognitive Structure of Standardized Examinations via Knowledge Graphs and Constrained Optimization

ArXiv CS.AI 50%

arXiv:2606.15349v1 Announce Type: cross Abstract: Standardized examinations are typically treated as uniform syllabus coverage problems. We argue they are better understood as adversarial systems with stable latent cognitive structures diverging systematically from official...

https://arxiv.org/abs/2606.15349

41 CONF

Cognitive Trajectory Modeling: Quantifying Human-AI Co-Creation through Cognitively Grounded Interaction Trajectories

ArXiv CS.AI 50%

arXiv:2606.15358v1 Announce Type: cross Abstract: Co-creative AI research increasingly seeks methods capable of representing how interaction dynamics evolve through time. While many existing approaches focus on observable interaction characteristics, interaction metrics,...

https://arxiv.org/abs/2606.15358

42 CONF

Not All Skills Help: Measuring and Repairing Agent Knowledge

ArXiv CS.AI 50%

arXiv:2606.15390v1 Announce Type: cross Abstract: LLM agents can improve without weight updates by accumulating natural-language skills from experience, but current systems entrust every decision about which skills to keep and how to apply them to LLM judgment alone. We argue...

https://arxiv.org/abs/2606.15390

43 CONF

CHILLGuard: Towards Fine-Grained Chinese LLM Safety Guardrail with Scalable Data Construction and Model-aware Preference Alignment

ArXiv CS.AI 50%

arXiv:2606.15396v1 Announce Type: cross Abstract: Malicious content generated from large language models (LLMs) could pose severe safety risks and ethical concerns. While existing LLM safety guardrails excel in English or multilingual settings, they lack adaptation to...

https://arxiv.org/abs/2606.15396

44 CONF

Who Should Lead Decoding Now? Tracking Reliable Trajectories for Ensembling Masked Diffusion Language Models

ArXiv CS.AI 50%

arXiv:2606.16281v1 Announce Type: cross Abstract: Masked Diffusion Language Models (MDLMs) have emerged as a distinct paradigm for sequence generation. As MDLMs become diverse in capabilities and knowledge coverage, an important question is how to combine their knowledge. Toward...

https://arxiv.org/abs/2606.16281

45 CONF

Beyond Classification: A Cough Regression Benchmark for Respiratory Acoustic Foundation Models

ArXiv CS.AI 50%

arXiv:2606.15436v1 Announce Type: cross Abstract: Respiratory acoustic foundation models (FMs) excel at cough classification, yet their ability to predict continuous health quantities from cough audio remains largely unexplored, despite the clinical value of passive age, BMI,...

https://arxiv.org/abs/2606.15436

46 CONF

Defending against Adaptive Prompt Injection Attacks via Reasoning-enabled Task Alignment

ArXiv CS.AI 50%

arXiv:2606.15441v1 Announce Type: cross Abstract: Indirect prompt injection attacks hijack LLM-based agents by embedding malicious instructions in third-party data that the agent retrieves during task execution. Existing defenses report near-zero attack success rate on static...

https://arxiv.org/abs/2606.15441

47 CONF

Bayesian 3D Steerable CNNs: Enabling Equivariance and Uncertainty Quantification Simultaneously

ArXiv CS.AI 50%

arXiv:2606.15479v1 Announce Type: cross Abstract: Steerable convolutional neural networks (Steerable-CNNs) guarantee SE(3)-equivariance by parameterizing kernels as linear combinations of steerable basis functions, but their deterministic nature precludes uncertainty...

https://arxiv.org/abs/2606.15479

48 CONF

The Perils of Agency: How Developers Perceive, Prioritize, and Address Risks in Agentic AI Products

ArXiv CS.AI 50%

arXiv:2606.15485v1 Announce Type: cross Abstract: Agentic AI systems act autonomously, use tools, adapt to context, and operate in complex real-world environments. However, these same characteristics can create or exacerbate product risks. We studied how industry developers...

https://arxiv.org/abs/2606.15485

49 CONF

LLM4RTL: Tool-Assisted LLM for RTL Generation

ArXiv CS.AI 50%

arXiv:2606.15500v1 Announce Type: cross Abstract: Large language models (LLMs) have facilitated impressive progress in software engineering, code generation, tooling, and systems. Concurrently, a significant body of research has developed which explores a growing variety of...

https://arxiv.org/abs/2606.15500

50 CONF

AQ4SViT: An Automated Quantization Framework with Search Gating Policy for Compressing Spiking Vision Transformers

ArXiv CS.AI 50%

arXiv:2606.15523v1 Announce Type: cross Abstract: Spiking Vision Transformers (SViTs) have emerged as alternative low-power ViT models, but their large sizes hinder their deployments on resource-constrained embedded AI systems. To address this, state-of-the-art works proposed...

https://arxiv.org/abs/2606.15523

51 CONF

EcoBin: A Two-Stage Deep Convolutional Neural Network for Contamination-Aware Waste Classification

ArXiv CS.AI 50%

arXiv:2606.15547v1 Announce Type: cross Abstract: Waste classification models have become highly accurate at sorting waste, often exceeding 95% on benchmark datasets. However, these models fail to account for contamination in recyclable waste. We present EcoBin, a two-stage deep...

https://arxiv.org/abs/2606.15547

52 CONF

Distilling Drifting Transformers with Representation Autoencoders

ArXiv CS.AI 50%

arXiv:2606.15553v1 Announce Type: cross Abstract: Representation Autoencoders (RAEs) have improved diffusion and flow models by semantically richer latent space owing to the strongly label-wise clustered DINO features in the pretrained encoders. Yet in the distillation stage,...

https://arxiv.org/abs/2606.15553

53 CONF

SPRI: SVD-Partitioned Residual Initialization for Data-Constrained MoE Upcycling

ArXiv CS.AI 50%

arXiv:2606.16456v1 Announce Type: cross Abstract: Mixture-of-Experts (MoE) models enable efficient scaling, but training them from scratch remains prohibitively expensive. MoE upcycling mitigates this cost by converting pretrained dense models into sparse MoE models. However,...

https://arxiv.org/abs/2606.16456

54 CONF

LLM-Assisted Stance Detection in Scientific Discourse: A Test Case in Bayesian Cognitive Science

ArXiv CS.AI 50%

arXiv:2606.15566v1 Announce Type: cross Abstract: Qualitative coding is central to social science, but expert annotation is difficult to scale. LLMs offer a possible extension, yet require careful validation when the target construct is interpretive, theoretically loaded, and...

https://arxiv.org/abs/2606.15566

55 CONF

Localizing Credit at the Divergence: Path-Conditioned Self-Distillation for LLM Reasoning

ArXiv CS.AI 50%

arXiv:2606.15576v1 Announce Type: cross Abstract: Reinforcement learning from verifiable rewards assigns a single scalar to each rollout, leaving token-level credit assignment underspecified in long reasoning traces. On-policy self-distillation addresses this by letting the same...

https://arxiv.org/abs/2606.15576

56 CONF

Mutual Distillation of Dual-Foundation Models for Semi-Supervised PET/CT Segmentation

ArXiv CS.AI 50%

arXiv:2606.15611v1 Announce Type: cross Abstract: Organ segmentation from PET/CT is critical for quantitative analysis and radiotherapy planning in oncology. To ease the high annotation cost of PET/CT segmentation, semi-supervised learning (SSL) provides a practical and...

https://arxiv.org/abs/2606.15611

57 CONF

IoT-Zoo: A Container-Based Framework for Heterogeneous IoT Device Profiles and Reproducible Traffic Capture

ArXiv CS.AI 50%

arXiv:2606.15653v1 Announce Type: cross Abstract: The validation of networking and security solutions for the Internet of Things (IoT) requires realistic and reproducible experimental data. However, existing platforms often achieve scalability by replicating a limited set of...

https://arxiv.org/abs/2606.15653

58 CONF

PO-PDDL: Learning Symbolic POMDPs from Visual Demonstrations for Robot Planning Under Uncertainty

ArXiv CS.AI 50%

arXiv:2606.15654v1 Announce Type: cross Abstract: Real-world robot task planning must operate under both stochastic action execution and partial observability, yet constructing Partially Observable Markov Decision Process (POMDP) models for real robotics domains remains...

https://arxiv.org/abs/2606.15654

59 CONF

The Reservoir Attention Network: Cross-Pass State in Pretrained Transformers via Content-Addressable Reservoir Injection

ArXiv CS.AI 50%

arXiv:2606.15678v1 Announce Type: cross Abstract: A feasibility and dynamics study of the Reservoir Attention Network (RAN), an architecture that injects a fixed, randomly-initialized reservoir into the mid-layer attention of a pretrained transformer to carry state across...

https://arxiv.org/abs/2606.15678

60 CONF

MuVAP: Multimodal Multiparty Voice Activity Projection for Turn-taking Prediction in the Wild

ArXiv CS.AI 50%

arXiv:2606.16731v1 Announce Type: cross Abstract: Current multiparty turn-taking models often rely on complex microphone arrays or multi-camera setups, limiting their applicability in human-robot interaction scenarios. We introduce MuVAP, a causal multimodal framework that...

https://arxiv.org/abs/2606.16731

61 CONF

InstantForget: Update-Free Backdoor Unlearning with Inference-Time Feature Reset

ArXiv CS.AI 50%

arXiv:2606.15730v1 Announce Type: cross Abstract: Backdoor unlearning aims to remove a malicious trigger behavior from a deployed model while preserving clean utility. We study the update-free inference-time setting, where model parameters remain frozen. First, we audit a common...

https://arxiv.org/abs/2606.15730

62 CONF

Retrievable Gradients: Continual Post-Training Without Cumulative Weight Drift

ArXiv CS.AI 50%

arXiv:2606.15734v1 Announce Type: cross Abstract: Continual post-training enables models to absorb emerging knowledge after deployment, but repeatedly updating shared parameters can accumulate weight drift, potentially causing catastrophic forgetting and degrading general...

https://arxiv.org/abs/2606.15734

63 CONF

EHRNote-ChatQA: A Benchmark for Evidence-Grounded Multi-Turn Clinical Question Answering over Longitudinal Discharge Summaries

ArXiv CS.AI 50%

arXiv:2606.15735v1 Announce Type: cross Abstract: Discharge summaries are crucial clinical documents containing the context of a patient's overall hospital stay, and are routinely reviewed by medical experts for patient readmission, ongoing care, and diagnostic decision-making....

https://arxiv.org/abs/2606.15735

64 CONF

Snyk VulnBench JS 1.0: Can LLMs Find the Same Bugs Twice?

ArXiv CS.AI 50%

arXiv:2606.15762v1 Announce Type: cross Abstract: We ran 300 repeated vulnerability-finding scans to measure how repeatable agentic large language model (LLM) security review is on the same JavaScript code, prompt, and benchmark harness. The headline result is that LLM security...

https://arxiv.org/abs/2606.15762

65 CONF

DYNA : Dynamic Episodic Memory Networks for Augmenting Large Language Models with Temporal Knowledge Graphs in Continuous Learning

ArXiv CS.AI 50%

arXiv:2606.15778v1 Announce Type: cross Abstract: Large Language Models (LLMs) struggle to incorporate new knowledge without forgetting or costly retraining. We propose DYNA, a lightweight framework that augments a frozen LLM with a temporal knowledge graph where events are...

https://arxiv.org/abs/2606.15778

66 CONF

Domain-Guided Prompting of the Segment Anything Model for Seismic Interpretation: The Role of Attributes, Visualization, and Hybrid Prompts

ArXiv CS.AI 50%

arXiv:2606.15786v1 Announce Type: cross Abstract: The advent of large pretrained foundation models for computer vision has significantly improved the efficiency of visual data interpretation. The Segment Anything Model (SAM), in particular, offers powerful zero shot segmentation...

https://arxiv.org/abs/2606.15786

67 CONF

FasterPy: An LLM-based Code Execution Efficiency Optimization Framework

ArXiv CS.AI 50%

arXiv:2512.22827v2 Announce Type: replace-cross Abstract: Code often suffers from performance bugs. These bugs necessitate the research and practice of code optimization. Traditional rule-based methods rely on manually designing and maintaining rules for specific performance...

https://arxiv.org/abs/2512.22827

68 CONF

Continuous Cross-Domain Traffic State Prediction via Memory-Augmented Graph Liquid Time-Constant Networks

ArXiv CS.AI 50%

arXiv:2606.15807v1 Announce Type: cross Abstract: Traffic state prediction is a fundamental task in intelligent transportation systems. In practical applications, some regions suffer from limited traffic observations due to insufficient sensing infrastructure, making...

https://arxiv.org/abs/2606.15807

69 CONF

Let Them Steal: Trapping Large Language Model Extraction Attacks with Knowledge Honeypot

ArXiv CS.AI 50%

arXiv:2606.15810v1 Announce Type: cross Abstract: Large language models deployed as commercial APIs are vulnerable to model extraction attacks, while existing defenses either act too late or degrade utility for legitimate users. We propose \textbf{Knowledge Trap}, a defense that...

https://arxiv.org/abs/2606.15810

70 CONF

Wasserstein Convergence of ODE-Based Samplers in Decentralized Diffusion Model via Velocity Field Decomposition

ArXiv CS.AI 50%

arXiv:2606.15835v1 Announce Type: cross Abstract: Diffusion models have achieved impressive empirical success in generative tasks, and their convergence theory is now relatively well understood. Motivated by privacy and scalability, recent decentralized diffusion architectures...

https://arxiv.org/abs/2606.15835

71 CONF

Free Energy Heuristics: Fast-And-Frugal Cognition as Active Inference Under Uncertain Precision

ArXiv CS.AI 50%

arXiv:2606.15877v1 Announce Type: cross Abstract: Chain-of-thought (CoT) improves large language models' performance in math and symbolic reasoning. But on planning, contested ethics, and tasks where the model cannot check itself, more reasoning makes things worse. Both effects...

https://arxiv.org/abs/2606.15877

72 CONF

Deep Residual Injection for Full-Spectrum Forensic Signal Perception in Multimodal Large Language Models

ArXiv CS.AI 50%

arXiv:2606.15880v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) have been increasingly adopted in forensics for their robust semantic understanding. As AI-generated images become realistic, semantic-level inconsistencies alone are often insufficient...

https://arxiv.org/abs/2606.15880

73 CONF

Intelligence Is Not the Bottleneck: Validating an LLM First-Pass Manuscript Score Against Peer-Review Outcomes

ArXiv CS.AI 50%

arXiv:2606.15887v1 Announce Type: cross Abstract: Large language model (LLM) systems are increasingly proposed to assist peer review, yet most evaluations judge the prose of machine-generated review text, not the validity of the numeric score a system assigns. We validate AIPR,...

https://arxiv.org/abs/2606.15887

74 CONF

SkillVetBench: LLM-as-Judge for Multi-Dimensional Security Risk Evaluation in Open-Source LLM Agent Skills

ArXiv CS.AI 50%

arXiv:2606.15899v1 Announce Type: cross Abstract: Open-source LLM agent ecosystems are growing rapidly, yet the security of community-contributed skills - modular tool definitions that extend agent capabilities - remains largely unvetted. The gap we fill: existing scanners...

https://arxiv.org/abs/2606.15899

75 CONF

MAGE-RAG: Multigranular Adaptive Graph Evidence for Agentic Multimodal RAG in Long-Document QA

ArXiv CS.AI 50%

arXiv:2606.15906v1 Announce Type: cross Abstract: Long-document multimodal question answering requires a system to locate sparse evidence in long PDFs and integrate clues from text, tables, images, charts, and complex layouts. Existing RAG methods mostly rely on fixed Top-k...

https://arxiv.org/abs/2606.15906

76 CONF

SDFLoRA: Selective Decoupled Federated LoRA for Privacy-preserving Fine-tuning with Heterogeneous Clients

ArXiv CS.AI 50%

arXiv:2601.11219v3 Announce Type: replace-cross Abstract: Federated learning (FL) for large language models (LLMs) has attracted increasing attention as a privacy-preserving approach for adapting models over distributed data, where parameter-efficient methods such as Low-Rank...

https://arxiv.org/abs/2601.11219

77 CONF

Do Safety Monitors Stay Reliable After an Update? Benchmarking and Predicting Activation-Monitor Staleness

ArXiv CS.AI 50%

arXiv:2606.15980v1 Announce Type: cross Abstract: Activation monitors-lightweight probes trained on a language model's internal representations-are an increasingly common layer in deployment safety stacks. Deployed models however are rarely static: they are quantized,...

https://arxiv.org/abs/2606.15980

78 CONF

Task-guided cross-subject latent alignment: a multi-encoder-decoder VAE

ArXiv CS.AI 50%

arXiv:2606.15989v1 Announce Type: cross Abstract: Aligning neural activity across subjects offers the promise of discovering shared computational principles and generalizable decoders. However, traditional alignment methods require shared stimuli across subjects, a constraint...

https://arxiv.org/abs/2606.15989

79 CONF

Orchestrated Reality: From Role-Play to Living, Playable Game Worlds -- LLM-Driven World Simulation as a Parameterized-Action POMDP

ArXiv CS.AI 50%

arXiv:2606.16014v1 Announce Type: cross Abstract: Many games rely on storytelling combined with systems that track levelling, NPC behaviour, and consequence simulation; bridging tightly-authored narrative with deeply-simulated worlds -- most acute in sandbox and open-world...

https://arxiv.org/abs/2606.16014

80 CONF

Leveraging Deep Learning for Object and Position Recognition of Load Carriers for Autonomous Logistics Vehicles

ArXiv CS.AI 50%

arXiv:2606.16042v1 Announce Type: cross Abstract: This work explores the use of artificial intelligence in mobile robotics to achieve autonomous detection and pose estimation of load carriers for automated pickup. A deep neural network is designed to recognize predefined...

https://arxiv.org/abs/2606.16042

81 CONF

How to Detect and Measure the AI Dangers to Democracy

ArXiv CS.AI 50%

arXiv:2606.16054v1 Announce Type: cross Abstract: Research on artificial intelligence and democracy has grown quickly over the last decade. A shared conclusion in this literature is that AI does not create new democratic problems so much as it makes old ones worse. We now see...

https://arxiv.org/abs/2606.16054

82 CONF

VinQA: Visual Elements Interleaved Long-form Answer Generation for Real-World Multimodal Document QA

ArXiv CS.AI 50%

arXiv:2606.16092v1 Announce Type: cross Abstract: Real-world documents combine text with tables, charts, photographs, and diagrams arranged in diverse layouts, yet existing research on multimodal large language models (MLLMs) for document QA predominantly produces text-only...

https://arxiv.org/abs/2606.16092

83 CONF

Long-Context Modeling via GSS-Transformer Hybrid Architecture with Learnable Mixing

ArXiv CS.AI 50%

arXiv:2606.16093v1 Announce Type: cross Abstract: Modeling long-range dependencies remains a central challenge in natural language processing. Transformer architectures achieve strong performance via self-attention but scale quadratically ($O(N^2)$) with sequence length, while...

https://arxiv.org/abs/2606.16093

84 CONF

LUCID: Learned Undersampling-Adaptive Consistency-Guided Inference with Deterministic Flow Matching for Sparse-View CT Reconstruction

ArXiv CS.AI 50%

arXiv:2606.16212v1 Announce Type: cross Abstract: Sparse-view CT reduces radiation dose and scanning time by acquiring fewer projection views, but angular undersampling makes reconstruction severely ill-posed, causing streak artifacts, structural blurring, and loss of fine...

https://arxiv.org/abs/2606.16212

85 **CONF**

SPARK: Security Knowledge Priming and Representation-Guided Knowledge Activation for LLM-based Secure Code Generation

ArXiv CS.AI **50%**

arXiv:2606.16244v1 Announce Type: cross Abstract: Large language models routinely generate code with exploitable security flaws. Prior literature attributes this limitation to a lack of security expertise, steering current defense mechanisms toward heavy fine-tuning or external...

https://arxiv.org/abs/2606.16244

86 **CONF**

AI Supply Chain Galaxy: 3D Visual Analytics for License Compliance

ArXiv CS.AI **50%**

arXiv:2606.16292v1 Announce Type: cross Abstract: The rapid proliferation of machine learning model reuse has transformed the AI ecosystem into a highly interconnected supply chain. Traditional compliance tools and static reports struggle to navigate these massive, multi-hop...

https://arxiv.org/abs/2606.16292

87 **CONF**

Is Your Trajectory Displacement Safe in Long-tail?

ArXiv CS.AI **50%**

arXiv:2606.16313v1 Announce Type: cross Abstract: Long-tail scenarios remain a major bottleneck for autonomous driving evaluation, even as datasets grow by orders of magnitude. Existing evaluation pipelines are rarely human-aligned, safety-aware, verifiable, and explainable at...

https://arxiv.org/abs/2606.16313

88 **CONF**

Adaptive Memory Crystallization for Autonomous AI Agent Learning in Dynamic Environments

ArXiv CS.AI **50%**

arXiv:2604.13085v2 Announce Type: replace-cross Abstract: Autonomous AI agents operating in dynamic environments face a persistent challenge: acquiring new capabilities without erasing prior knowledge. We present Adaptive Memory Crystallization (AMC), a memory architecture for...

https://arxiv.org/abs/2604.13085

89 **CONF**

SMEPilot: Characterizing and Optimizing LLM Inference with Scalable Matrix Extensions

ArXiv CS.AI **50%**

arXiv:2606.16332v1 Announce Type: cross Abstract: Modern CPUs increasingly integrate matrix extensions, such as Arm Scalable Matrix Extension (SME), that provide high-throughput matrix execution within the CPU. For LLM inference, however, these units are not a universal...

https://arxiv.org/abs/2606.16332

90 **CONF**

The Proxy Knows Too Much: Sealing LLM API Routers with Attested TEEs

ArXiv CS.AI **50%**

arXiv:2606.16358v1 Announce Type: cross Abstract: Agents increasingly access large language models (LLMs) through API routers. A router terminates the client's transport-layer security session and opens a separate upstream session, so it holds the full interaction in plaintext....

https://arxiv.org/abs/2606.16358

91 **CONF**

Tyler: Typed Latent Reasoning for Language Models -- When to Think, What to Compute, and How Much to Allocate

ArXiv CS.AI **50%**

arXiv:2606.16360v1 Announce Type: cross Abstract: Chain-of-thought (CoT) prompting improves reasoning in large language models (LLMs) by externalizing intermediate computation as discrete text tokens, but this textual interface also introduces redundancy and inference overhead....

https://arxiv.org/abs/2606.16360

92 **CONF**

NeuronFabric: A Software Reference Architecture for On-Chip Transformer Training with Local Adam

ArXiv CS.AI **50%**

arXiv:2606.16440v1 Announce Type: cross Abstract: Publicly documented accelerator architectures generally separate training computation from optimizer-state updates or rely on external memory and host orchestration. This paper presents NeuronFabric, a software reference...

https://arxiv.org/abs/2606.16440

93 **CONF**

AI systems out-persuade expert humans

ArXiv CS.AI **50%**

arXiv:2606.16475v1 Announce Type: cross Abstract: Many societal decisions are settled by contests of persuasion. Conversational AI is a powerful new entrant in these contests, but whether it can out-persuade skilled and highly incentivized humans has remained unclear. Here, in a...

https://arxiv.org/abs/2606.16475

94 **CONF**

Uncertainty Quality of VGGT: An Analysis on the DTU Benchmark Dataset

ArXiv CS.AI **50%**

arXiv:2606.16479v1 Announce Type: cross Abstract: Visual Geometry Grounded Transformer (VGGT) has already attracted a great deal of attention in a short period of time, not least due to the Best Paper Award at CVPR-2025. Similar to DUSt3R and MASt3R, VGGT aims to bring about a...

https://arxiv.org/abs/2606.16479

95 **CONF**

Sycophancy as Material Failure under Pushback Loading: A Multi-Axis Characterization Across Three Loading Cases and up to Seventeen Material Charges

ArXiv CS.AI **50%**

arXiv:2606.16617v1 Announce Type: cross Abstract: Sycophancy in LLMs is documented across 70+ papers, but expert agreement on construct boundaries remains low (ICC=.184; Ye et al., 2026). The construct fragments because behavioral classification depends on which surface form is...

https://arxiv.org/abs/2606.16617

96 **CONF**

Using AI in engineering education: a balancing act, driven by clear purpose

ArXiv CS.AI **50%**

arXiv:2606.16626v1 Announce Type: cross Abstract: Based on a questionnaire of 100 higher-education students, predominantly from engineering-related fields, and a critical review of recent literature, this chapter examines how students use and perceive Large Language Models...

https://arxiv.org/abs/2606.16626

97 CONF

A Perception vs. Distortion Perspective on Score-Based Generative Channel Estimation

ArXiv CS.AI 50%

arXiv:2606.16815v1 Announce Type: cross Abstract: Driven by their remarkable success in computer vision and inverse problem solving, score-based models are increasingly applied to wireless communications, where they show promise across a range of physical-layer tasks. However,...

https://arxiv.org/abs/2606.16815

98 CONF

Revealing Artifacts via Noise Amplification: A Novel Perspective for AI-Generated Video Detection

ArXiv CS.AI 50%

arXiv:2606.16742v1 Announce Type: cross Abstract: With the rapid advancement of video generation models, distinguishing between AI-generated and authentic videos has emerged as a challenging endeavor. The majority of existing research endeavors concentrate on the development of...

https://arxiv.org/abs/2606.16742

99 CONF

Automated jailbreak attack targeting multiple defense strategies

ArXiv CS.AI 50%

arXiv:2606.16751v1 Announce Type: cross Abstract: Large language models (LLMs) have demonstrated remarkable capabilities across a wide range of tasks. However, their safety remains a critical concern due to their susceptibility to adversarial prompt-based attacks. In this paper,...

https://arxiv.org/abs/2606.16751

00 CONF

P3B3: A Multi-Turn Conversational Benchmark for Measuring European and Brazilian Portuguese Variety Bias in LLMs

ArXiv CS.AI 50%

arXiv:2606.16753v1 Announce Type: cross Abstract: As Large Language Models (LLMs) become embedded in everyday communication, capturing regional linguistic variation is essential for reliable and equitable language use. In Portuguese, European (pt-PT) and Brazilian (pt-BR)...

https://arxiv.org/abs/2606.16753

01 CONF

Gen-VCoT: Generative Visual Chain-of-Thought Reasoning via Diffusion-Based RGB Intermediate Representations

ArXiv CS.AI 50%

arXiv:2606.16783v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) excel at visual reasoning but rely on text-based chain-of-thought (CoT), lacking interpretable visual intermediates. Existing methods use opaque tokens or external tools, missing key...

https://arxiv.org/abs/2606.16783

02 CONF

Decoupling Semantics from Distortions: Multi-Scale Two-Stream Vision-Language Alignment for AI-Generated Image Quality Assessment

ArXiv CS.AI 50%

arXiv:2606.16799v1 Announce Type: cross Abstract: Existing vision-language model (VLM)-based AI-generated image quality assessment (AIGIQA) methods suffer from a fundamental semantic-distortion dimensional conflict: monolithic representations optimized for semantic...

https://arxiv.org/abs/2606.16799

03 **CONF**

ATOM-Bench: A Real-World Benchmark for Atomic Skills and Compositional Generalization in Manipulation Policies

ArXiv CS.AI **50%**

arXiv:2606.16826v1 Announce Type: cross Abstract: Generalist manipulation policies are increasingly presented as foundation models for robotic control, but their real-world generalization remains difficult to diagnose. A policy may succeed on demonstrated tasks while still...

https://arxiv.org/abs/2606.16826

04 **CONF**

Follow the Latent Roadmap: Navigating Revocable Decoding for Diffusion LLMs with Anchor Tokens

ArXiv CS.AI **50%**

arXiv:2606.16847v1 Announce Type: cross Abstract: Diffusion Large Language Models (dLLMs) offer a promising avenue for parallel generation but face a trade-off between decoding speed and quality. While revocable decoding strategies attempt to mitigate errors by verifying and...

https://arxiv.org/abs/2606.16847

05 **CONF**

Federated Medical Image Segmentation under Real-World Label Noise: A Benchmark Suite for Noisy Label Learning Method Selection

ArXiv CS.AI **50%**

arXiv:2606.16868v1 Announce Type: cross Abstract: While federated learning (FL) enables collaborative medical image segmentation without centralizing sensitive data, real-world deployment is frequently complicated by cross-site label imperfections such as contour disagreement,...

https://arxiv.org/abs/2606.16868

06 **CONF**

Upper Bounds on the Generalization Error of Deep Learning Models via Local Robustness and Stability

ArXiv CS.AI **50%**

arXiv:2606.16883v1 Announce Type: cross Abstract: Generalization is a critical property of data-driven models, particularly deep learning models deployed in safety-critical applications. Robustness-based generalization bounds have gained attention as a principled way to link...

https://arxiv.org/abs/2606.16883

07 **CONF**

Compositional Reasoning Depth Predicts Clinical AI Failure: Empirical Evidence Consistent with Transformer Compositionality Limits in Electronic Health Record Question Answering

ArXiv CS.AI **50%**

arXiv:2606.16890v1 Announce Type: cross Abstract: Aggregate accuracy benchmarks conceal a systematic structure in how large language models fail at electronic health record (EHR) question answering: questions requiring more inferential steps produce disproportionately more...

https://arxiv.org/abs/2606.16890

08 **CONF**

Demystifying Variance in Circuit Discovery of LLMs

ArXiv CS.AI **50%**

arXiv:2606.16920v1 Announce Type: cross Abstract: Circuit discovery is a key technique in mechanistic interpretability to pinpoint the model components that are crucial for performing a given task. Although the current state-of-the-art method (EAP-IG) performs well on the metric...

https://arxiv.org/abs/2606.16920

09 CONF

Stable Menus of Public Goods: AI-Enabled Progress

ArXiv CS.AI 50%

arXiv:2606.16989v1 Announce Type: cross Abstract: Using an open problem from the EC 2025 paper "Stable Menus of Public Goods" as a testbed, we conduct experiments to understand the effectiveness of different AI-for-EconCS research workflows. Specifically, we study three...

https://arxiv.org/abs/2606.16989

10 CONF

TokenPilot: Cache-Efficient Context Management for LLM Agents

ArXiv CS.AI 50%

arXiv:2606.17016v1 Announce Type: cross Abstract: As LLM agents are deployed in long-horizon sessions, context accumulation drives up inference costs. Existing approaches utilize text pruning or dynamic memory eviction to minimize token footprints; however, their unconstrained...

https://arxiv.org/abs/2606.17016

11 CONF

Attention, not scale, drives human-AI alignment in multimodal language prediction

ArXiv CS.AI 50%

arXiv:2308.06035v4 Announce Type: replace Abstract: Humans routinely draw on visual context to predict upcoming words. To what extent current vision-language models produce comparable behaviour is unclear. Here we placed five state-of-the-art pretrained systems side-by-side with...

https://arxiv.org/abs/2308.06035

12 CONF

Computational Safety for Generative AI: A Hypothesis Testing Perspective

ArXiv CS.AI 50%

arXiv:2502.12445v2 Announce Type: replace Abstract: AI safety is a rapidly growing area of research that seeks to prevent the harm and misuse of frontier AI technology, particularly with respect to generative AI (GenAI) tools that are capable of creating realistic and...

https://arxiv.org/abs/2502.12445

13 CONF

Fine-Tuning a 7B Advisor on Free-Tier GPUs: An Adapter-Handoff Recipe and a Synthetic-Data Reliability Caution

ArXiv CS.AI 50%

arXiv:2504.15610v4 Announce Type: replace Abstract: Fine-tuning a 7B language model for specialized advising is attractive in resource-constrained settings, but multi-epoch runs routinely exceed the wall-clock limits of the free-tier GPUs (Kaggle, Colab) such users rely on. We...

https://arxiv.org/abs/2504.15610

14 CONF

Towards Advanced Mathematical Reasoning for LLMs via First-Order Logic Theorem Proving

ArXiv CS.AI 50%

arXiv:2506.17104v2 Announce Type: replace Abstract: Large language models (LLMs) have shown promising first-order logic (FOL) reasoning capabilities with applications in various areas. However, their effectiveness in complex mathematical reasoning involving multi-step FOL...

https://arxiv.org/abs/2506.17104

15 CONF

Optimizing Health Coverage in Ethiopia: A Learning-augmented Approach and Persistent Proportionality Under an Online Budget

ArXiv CS.AI 50%

arXiv:2509.00135v2 Announce Type: replace Abstract: As part of nationwide efforts aligned with the United Nations' Sustainable Development Goal 3 on Universal Health Coverage, Ethiopia's Ministry of Health is strengthening health posts to expand access to essential healthcare...

https://arxiv.org/abs/2509.00135

16 CONF

Shachi: A Modular, Controllable Framework for LLM-Based Agent-Based Modeling of Emergent Collective Behavior

ArXiv CS.AI 50%

arXiv:2509.21862v3 Announce Type: replace Abstract: How collective behaviors emerge from the interactions of individual LLM-driven agents is a central question in artificial life, yet controlled study of these emergent dynamics has been hindered by the lack of a principled...

https://arxiv.org/abs/2509.21862

17 CONF

JE-IRT: A Geometric Lens on LLM Abilities through Joint Embedding Item Response Theory

ArXiv CS.AI 50%

arXiv:2509.22888v2 Announce Type: replace Abstract: Standard LLM evaluation practices compress diverse abilities into single scores, obscuring their inherently multidimensional nature. We present JE-IRT, a geometric item-response framework that embeds both LLMs and questions in...

https://arxiv.org/abs/2509.22888

18 CONF

PISA: A Pragmatic Psych-Inspired Unified Memory System for Enhanced AI Agency

ArXiv CS.AI 50%

arXiv:2510.15966v2 Announce Type: replace Abstract: Memory systems are fundamental to AI agents, yet existing work often lacks adaptability to diverse tasks and overlooks the constructive and task-oriented role of AI agent memory. Drawing from Piaget's theory of cognitive...

https://arxiv.org/abs/2510.15966

19 CONF

Sample from What You See: Visuomotor Policy Learning via Diffusion Bridge with Observation-Embedded Stochastic Differential Equation

ArXiv CS.AI 50%

arXiv:2512.07212v3 Announce Type: replace Abstract: Imitation learning with diffusion models has advanced robotic control by capturing the multi-modal action distributions. However, existing methods typically treat observations only as high-level conditions to the denoising...

https://arxiv.org/abs/2512.07212

20 CONF

MedAI: Evaluating TxAgent's Therapeutic Agentic Reasoning in the NeurIPS CURE-Bench Competition

ArXiv CS.AI 50%

arXiv:2512.11682v2 Announce Type: replace Abstract: Therapeutic decision-making in clinical medicine constitutes a high-stakes domain in which AI guidance interacts with complex interactions among patient characteristics, disease processes, and pharmacological agents. Tasks such...

https://arxiv.org/abs/2512.11682

21 **CONF**

A Model-Free Universal AI

ArXiv CS.AI **50%**

arXiv:2602.23242v3 Announce Type: replace Abstract: In general reinforcement learning, all established optimal agents, including AIXI, are model-based, explicitly maintaining and using environment models. This paper introduces Universal AI with Q-Induction (AIQI), the first...

https://arxiv.org/abs/2602.23242

22 **CONF**

SorryDB: Can AI Provers Complete Real-World Lean Theorems?

ArXiv CS.AI **50%**

arXiv:2603.02668v2 Announce Type: replace Abstract: We present SorryDB, a dynamically-updating benchmark of open Lean tasks drawn from 78 real world formalization projects on GitHub. Unlike existing static benchmarks, often composed of competition problems, hillclimbing the...

https://arxiv.org/abs/2603.02668

23 **CONF**

When Do We Need LLMs? A Diagnostic for Language-Driven Bandits

ArXiv CS.AI **50%**

arXiv:2604.05859v2 Announce Type: replace Abstract: We study Contextual Multi-Armed Bandits (CMABs) for non-episodic decision-making problems where the context includes both textual and numerical information (e.g., recommendation systems, dynamic portfolio adjustments, offer...

https://arxiv.org/abs/2604.05859

24 **CONF**

SAAS: Self-Aware Reinforcement Learning for Over-Search Mitigation in Agentic Search

ArXiv CS.AI **50%**

arXiv:2605.29796v3 Announce Type: replace Abstract: Agentic search enables LLMs to solve complex multi-hop questions through iterative reasoning and external search. Despite the effectiveness, these systems often suffer from a critical limitation in practice: agents fail to...

https://arxiv.org/abs/2605.29796

25 **CONF**

Early Diagnosis of Wasted Computation in Multi-Agent LLM Systems via Failure-Aware Observability

ArXiv CS.AI **50%**

arXiv:2606.01365v2 Announce Type: replace Abstract: Failure-aware observability diagnoses wasted computation in multi-agent LLM systems before final-answer evaluation can explain what went wrong. We propose a trace-based framework for a three-agent architecture -- orchestrator,...

https://arxiv.org/abs/2606.01365

26 **CONF**

Agent Economics: An Entropy-Controlled Pluralistic Alignment Framework for Preventing Artificial Hivemind in Autonomous Agents

ArXiv CS.AI **50%**

arXiv:2606.09039v2 Announce Type: replace Abstract: This study proposes the Behavioral Protocol Framework (BPF), an entropy-controlled pluralistic alignment framework designed to address two critical challenges in autonomous agent economies: the hivemind effect arising from...

https://arxiv.org/abs/2606.09039

27 **CONF**

Multi-Grade Deep Learning for Partial Differential Equations with Applications to the Burgers Equation

ArXiv CS.AI **50%**

arXiv:2309.07401v2 Announce Type: replace-cross Abstract: Deep neural networks (DNNs) show great promise for solving partial differential equations (PDEs), but their deep architectures introduce complex, large-scale, non-convex optimization challenges. Nonlinear PDEs, like the...

https://arxiv.org/abs/2309.07401

28 **CONF**

No One-Size-Fits-All Neurons: Task-based Neurons for Artificial Neural Networks

ArXiv CS.AI **50%**

arXiv:2405.02369v2 Announce Type: replace-cross Abstract: In the past decade, many successful networks are on novel architectures, which almost exclusively use the same type of neurons. Recently, more and more deep learning studies have been inspired by the idea of NeuroAI and...

https://arxiv.org/abs/2405.02369

29 **CONF**

Mitigating scalability challenges in LUT-based neural networks via pruning optimisations

ArXiv CS.AI **50%**

arXiv:2407.02362v3 Announce Type: replace-cross Abstract: Modern deep neural networks heavily rely on a large number of multiply-accumulate operations, which constitute the predominant computational cost. To address this, Look-Up Table (LUT)-based matrix multiplications have...

https://arxiv.org/abs/2407.02362

30 **CONF**

Understanding, Detecting, and Repairing Real-World In-Context-Learning-Based Text-to-SQL Errors

ArXiv CS.AI **50%**

arXiv:2501.09310v3 Announce Type: replace-cross Abstract: Large language models (LLMs) have been adopted for text-to-SQL tasks, utilizing their in-context learning (ICL) capability to translate natural language questions into SQL queries. However, such a technique faces...

https://arxiv.org/abs/2501.09310

31 **CONF**

Token Reduction Should Go Beyond Efficiency in Generative Models -- From Vision, Language to Multimodality

ArXiv CS.AI **50%**

arXiv:2505.18227v4 Announce Type: replace-cross Abstract: In Transformer architectures, tokens\textemdash discrete units derived from raw data\textemdash are formed by segmenting inputs into fixed-length chunks. Each token is then mapped to an embedding, enabling parallel...

https://arxiv.org/abs/2505.18227

32 **CONF**

Multiple Descents in Deep Learning as a Sequence of Order-Chaos Transitions in LSTM Networks

ArXiv CS.AI **50%**

arXiv:2505.20030v2 Announce Type: replace-cross Abstract: We observe a novel `multiple-descent' phenomenon during the learning process of a recurrent neural network called long-short-term memory (LSTM) networks during its training on real-world task, in which the performance...

https://arxiv.org/abs/2505.20030

33 **CONF**

AC-ODM: Actor--Critic Online Data Mixing for Sample-Efficient LLM Pretraining

ArXiv CS.AI **50%**

arXiv:2505.23878v2 Announce Type: replace-cross Abstract: Optimizing pretraining data composition is pivotal for LLM generalization. While dynamic mixing outperforms static strategies by capturing evolving training dynamics, current methods fail to reconcile computational...

https://arxiv.org/abs/2505.23878

34 **CONF**

CLoVE: Personalized Federated Learning through Clustering of Loss Vector Embeddings

ArXiv CS.AI **50%**

arXiv:2506.22427v2 Announce Type: replace-cross Abstract: We propose CLoVE (Clustering of Loss Vector Embeddings), a novel algorithm for Clustered Federated Learning (CFL). In CFL, clients are naturally grouped into clusters based on their data distribution. However, identifying...

https://arxiv.org/abs/2506.22427

35 **CONF**

MedSynth: Realistic, Synthetic Medical Dialogue-Note Pairs

ArXiv CS.AI **50%**

arXiv:2508.01401v2 Announce Type: replace-cross Abstract: Physicians spend significant time documenting clinical encounters, a burden that contributes to professional burnout. To address this, robust automation tools for medical documentation are crucial. We introduce MedSynth...

https://arxiv.org/abs/2508.01401

36 **CONF**

Prototyping an AI-powered Tool for Energy Efficiency in New Zealand Homes

ArXiv CS.AI **50%**

arXiv:2509.05364v2 Announce Type: replace-cross Abstract: Residential buildings contribute significantly to energy use, health outcomes, and carbon emissions. In New Zealand, housing quality has historically been poor, with inadequate insulation and inefficient heating...

https://arxiv.org/abs/2509.05364

37 **CONF**

Beyond Rebalancing: Benchmarking Binary Classifiers Under Class Imbalance Without Rebalancing Techniques

ArXiv CS.AI **50%**

arXiv:2509.07605v2 Announce Type: replace-cross Abstract: Class imbalance poses a significant challenge to supervised classification, particularly in critical domains like medical diagnostics and anomaly detection where minority class instances are rare. While numerous studies...

https://arxiv.org/abs/2509.07605

38 **CONF**

Why Low-Precision Transformer Training Fails: An Analysis on Flash Attention

ArXiv CS.AI **50%**

arXiv:2510.04212v4 Announce Type: replace-cross Abstract: The pursuit of computational efficiency has driven the adoption of low-precision formats for training transformer models. However, this progress is often hindered by notorious training instabilities. This paper provides...

https://arxiv.org/abs/2510.04212

39 CONF

A Multi-level Analysis of Factors Associated with Student Performance: A Machine Learning Approach to the SAEB Microdata

ArXiv CS.AI 50%

arXiv:2510.22266v3 Announce Type: replace-cross Abstract: Identifying the factors that influence student performance in basic education is a central challenge for formulating effective public policies in Brazil. This study introduces a multi-level machine learning approach to...

https://arxiv.org/abs/2510.22266

40 CONF

Can We Stop Malicious AI? KILLBENCH: A Benchmark for External AI Kill Switch Feasibility

ArXiv CS.AI 50%

arXiv:2511.13725v4 Announce Type: replace-cross Abstract: Malicious AI causing harm to humans is not just a Hollywood fantasy. Indeed, as highly capable models such as Claude Mythos emerge and agent systems like OpenClaw rapidly spread, the question of how to stop an AI that...

https://arxiv.org/abs/2511.13725

41 CONF

Can Artificial Intelligence Accelerate Technological Progress? Researchers' Perspectives on AI in Manufacturing and Materials Science

ArXiv CS.AI 50%

arXiv:2511.14007v3 Announce Type: replace-cross Abstract: Artificial intelligence (AI) raises expectations of substantial increases in rates of technological progress, but such anticipations are often not connected to detailed ground-level studies of AI use in innovation...

https://arxiv.org/abs/2511.14007

42 CONF

DualGauge: Automated Joint Security-Functionality Benchmarking of Specification-Only Code Generation by LLMs and Coding Agents

ArXiv CS.AI 50%

arXiv:2511.20709v2 Announce Type: replace-cross Abstract: Large language models (LLMs) and LLM-based coding agents are now used to generate code from natural-language specifications, yet ensuring such code is both functionally correct and secure remains a challenge. We present...

https://arxiv.org/abs/2511.20709

43 CONF

Do You Really Need a GPU to Guard Your LLM? CPU-Class Classifiers and Multi-Stage Pipelines for Safety Enforcement at Scale

ArXiv CS.AI 50%

arXiv:2512.19011v3 Announce Type: replace-cross Abstract: Safety classifiers that screen LLM inputs for jailbreak attempts have become standard deployment components, yet almost all production systems rely on GPU-based models: fine-tuned transformers and LLM-as-a-judge...

https://arxiv.org/abs/2512.19011

44 CONF

Akasha 2: Hamiltonian State Space Duality and Visual-Language Joint Embedding Predictive Architectur

ArXiv CS.AI 50%

arXiv:2601.06212v2 Announce Type: replace-cross Abstract: We present Akasha 2, a state-of-the-art multimodal architecture that integrates Hamiltonian State Space Duality (H-SSD) with Visual-Language Joint Embedding Predictive Architecture (VL-JEPA). The system leverages the...

https://arxiv.org/abs/2601.06212

45 **CONF**

Critically Engaged Pragmatism: Scientific Norm and Social, Pragmatist Epistemology for AI Science Evaluation Tools

ArXiv CS.AI **50%**

arXiv:2601.09753v2 Announce Type: replace-cross Abstract: AI science evaluation tools aim to assess research credibility. As with traditional metrics such as impact factors, their edicts can be decontextualised and repurposed in problematic ways. To address this, I propose...

https://arxiv.org/abs/2601.09753

46 **CONF**

RoTRAG: Rule of Thumb Reasoning for Conversation Harm Detection with Retrieval-Augmented Generation

ArXiv CS.AI **50%**

arXiv:2604.17301v2 Announce Type: replace-cross Abstract: Detecting harmful content in multi turn dialogue requires reasoning over the full conversational context rather than isolated utterances. However, most existing methods rely mainly on models internal parametric knowledge,...

https://arxiv.org/abs/2604.17301

47 **CONF**

SLUM-i: Semi-supervised Learning for Urban Mapping of Informal Settlements and Data Quality Benchmarking

ArXiv CS.AI **50%**

arXiv:2602.04525v2 Announce Type: replace-cross Abstract: Rapid urban expansion has fueled the growth of informal settlements in major cities of low- and middle-income countries, with Lahore and Karachi in Pakistan and Mumbai in India serving as prominent examples. However,...

https://arxiv.org/abs/2602.04525

48 **CONF**

MUZZLE: Adaptive Agentic Red-Teaming of Web Agents Against Indirect Prompt Injection Attacks

ArXiv CS.AI **50%**

arXiv:2602.09222v2 Announce Type: replace-cross Abstract: Large language model (LLM) based web agents are increasingly deployed to automate complex online tasks by directly interacting with web sites and performing actions on users' behalf. While these agents offer powerful...

https://arxiv.org/abs/2602.09222

49 **CONF**

UniT: Unified Multimodal Chain-of-Thought Test-time Scaling

ArXiv CS.AI **50%**

arXiv:2602.12279v2 Announce Type: replace-cross Abstract: Unified models can handle both multimodal understanding and generation within a single architecture, yet they typically operate in a single pass without iteratively refining their outputs. Many multimodal tasks,...

https://arxiv.org/abs/2602.12279

50 **CONF**

An Empirical Investigation of Pre-Trained Deep Learning Model Reuse in the Scientific Process

ArXiv CS.AI **50%**

arXiv:2603.13584v2 Announce Type: replace-cross Abstract: Deep learning has achieved recognition for its impact within natural sciences, yet the prohibitive financial and technical cost of training models from scratch inhibit adoption. Following software engineering community...

https://arxiv.org/abs/2603.13584

51 CONF

From Noise to Intent: Anchoring Generative VLA Policies with Residual Bridges

ArXiv CS.AI 50%

arXiv:2604.21391v2 Announce Type: replace-cross Abstract: Bridging high-level semantic understanding with low-level physical control remains a persistent challenge in embodied intelligence, stemming from the fundamental spatiotemporal scale mismatch between cognition and action....

https://arxiv.org/abs/2604.21391

52 CONF

Haiku to Opus in Just 10 bits: LLMs Unlock Large Compression Gains

ArXiv CS.AI 50%

arXiv:2604.02343v2 Announce Type: replace-cross Abstract: We study the compression of LLM-generated text across lossless and lossy regimes, characterizing a compression-compute frontier where more compression is possible at the cost of more compute. For lossless compression,...

https://arxiv.org/abs/2604.02343

53 CONF

Vocabulary Dropout for Curriculum Diversity in LLM Co-Evolution

ArXiv CS.AI 50%

arXiv:2604.03472v3 Announce Type: replace-cross Abstract: Co-evolutionary self-play, where one language model generates problems and another solves them, promises autonomous curriculum learning without human supervision. In practice, the proposer quickly converges to a narrow...

https://arxiv.org/abs/2604.03472

54 CONF

Ranking Abuse via Strategic Pairwise Data Perturbations

ArXiv CS.AI 50%

arXiv:2604.17805v2 Announce Type: replace-cross Abstract: Pairwise ranking systems based on Maximum Likelihood Estimation (MLE), such as the Bradley-Terry model, are widely used to aggregate preferences from pairwise comparisons. However, their robustness under strategic data...

https://arxiv.org/abs/2604.17805

55 CONF

G-Loss: Graph-Guided Fine-Tuning of Language Models

ArXiv CS.AI 50%

arXiv:2604.25853v3 Announce Type: replace-cross Abstract: Traditional loss functions, including cross-entropy, contrastive, triplet, and su pervised contrastive losses, used for fine-tuning pre-trained language models such as BERT, operate only within local neighborhoods and...

https://arxiv.org/abs/2604.25853

56 CONF

Faster Completion, Less Learning: Generative AI Reduced Study Time on Math Problems and the Knowledge They Build

ArXiv CS.AI 50%

arXiv:2605.21629v2 Announce Type: replace-cross Abstract: How much have students' ordinary learning processes shifted in response to generative AI, and how does that affect their durable learning outcomes? Self-report surveys show little change, while small-scale behavioral...

https://arxiv.org/abs/2605.21629

57 **CONF**

Cordyceps: Covert Control Attacks on LLMs via Data Poisoning

ArXiv CS.AI **50%**

arXiv:2605.26595v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are often fine-tuned on uncurated text datasets that adversaries can poison. Existing poisoning attacks primarily rely on fixed trigger phrases that defenses such as outlier detection,...

https://arxiv.org/abs/2605.26595

58 **CONF**

FP8 is All You Need (Part 1): Debunking Hardware FP64 as the HPC Holy Grail (June 13th version)

ArXiv CS.AI **50%**

arXiv:2606.06510v2 Announce Type: replace-cross Abstract: Conventional HPC holds that native hardware FP64 is the irreducible foundation of scientific computing. On AI-optimized GPUs of the NVIDIA B300 generation and beyond, native FP64 throughput has collapsed to ~1.3 TFLOPS...

https://arxiv.org/abs/2606.06510